

Potter Engineering Center Renovation Study
Part 2 - Food Service Options

Presented by American Structurepoint in association with
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The A.A. Potter Center is a classic example of a 1970's institutional building that could easily be dismissed.

From the exterior, large expanses of unbroken brick façade, the unyielding repetition of precast panels create a bland, dated, uninspiring image. On the interior, the triple combination of closed offices lining the perimeter, small deeply shaded windows, and square building footprint combine to create a severe lack of interior daylight. In its current state, it is not a structure fit to inspire students or staff.

These flaws, however, are not impossible to overcome. Moreover, the Potter Center has more qualities in its favor than it has negatives. Though unattractive, the exterior skin is in good condition and structurally sound. The structure itself is a concrete waffle slab system, which is not only in good condition but could be quite attractive if exposed. Floor-to-floor heights are generous. Its greatest asset though, is its location. The Potter Center sits in at the heart of campus, immediately adjacent to the Wilmeth Active Learning Center and Centennial Mall and a stone's throw from the Purdue Mall.

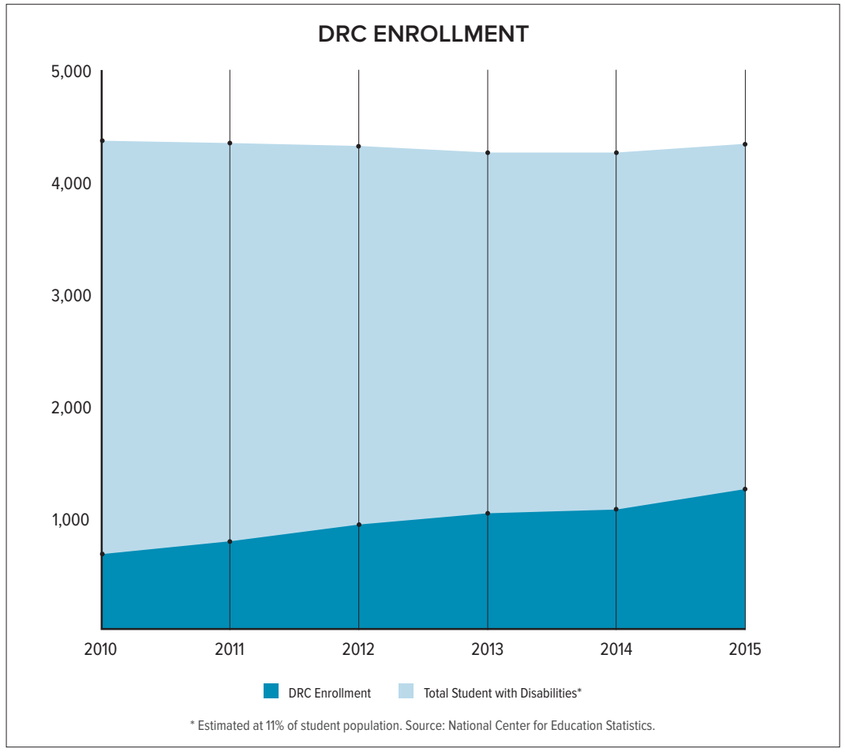
The Potter Center is the definitive example of a building in need of a renovation. Done thoughtfully, a renovation could overcome its current bad qualities and expand on its assets, easily breathing new life into an ideally located building.

PROGRAM

The programs/departments included in this study can generally be combined into four synergistic groups.

1. DISABILITY RESOURCE CENTER, ASSISTIVE TECHNOLOGY CENTER, INSTRUCTIONAL DATA PROCESSING, ONLINE COMPLIANCE

The Disability Resource Center is currently located on the eighth floor of Young Hall. As a department whose sole purpose is serving students with disabilities, this is not an ideal location. This study proposes to move the DRC into a more accessible, open, and welcoming first floor location. Additionally, this study proposes to expand the existing DRC program, which has expanded 85.5% over the last five years. While growing, these numbers appear to actually under-represent the number of students with disabilities. This number is expected to continue to increase as society continues to improve its understanding of disability as another form of diversity, civil rights, and social justice, allowing students to overcome the societal stigma of disability.



In addition to the DRC itself, the study proposes to co-locate several departments alongside the DRC to maximize efficiency, synergies, and productivity among those departments. The Assistive Technology Center provides students with disabilities the accommodations to do schoolwork, much in the same way the DRC currently provides students with disabilities the accommodations to take tests. This department has a very strong relationship and similar mission as the Alternate Format Production unit of the DRC.

Instructional Data Processing provides test scoring and evaluation services. IDP manages course evaluations, test proctoring services, Scantron grading, specialized gradebook work for certain departments, and Scantron scoring in support of research. The DRC's accommodated testing function primarily supports paper and pencil testing and is thus a major client of the IDP.

Online Compliance is a new unit, housed under Digital Education and led by an Associate Director of Digital Education. This department will ensure accommodation of digital information in various digital assets. Much of the work of this unit will support efforts of the DRC, although Digital Education will be another major user. The Associate Director position is funded and was recently filled. The remaining positions are yet to be funded. Therefore, these staff are new and not connected with the existing units.

2. CENTRAL TESTING CENTER (COMBINED DIGITAL EDUCATION TESTING CENTER, DRC ACCOMMODATED TESTING, FOREIGN LANGUAGE TESTING, PURDUE TESTING CENTER)

Rationale for Creation of a Centralized Testing Center

The Central Testing Center does not currently exist on campus. The University currently has several testing units that simultaneously provide similar testing services. These are the Disability Resource Center, Accommodated Testing, Purdue Testing Center, Digital Education Testing Center, and Foreign Language Testing. Each of these units has a small staff and small facility.

Because of the dispersion of testing programs, instructors must work with multiple offices for testing. A single exam might require working with the DRC for accommodations and perhaps an alternative format Instructional Data Processing (IDP) for proctoring and/or Scantrons, as well as Digital Education for online exams. There is currently no coordination among these different units.

Similarly, students work with multiple testing offices scattered across the campus. Students taking online courses may take their exam with Digital Education's testing unit. Students with disabilities may have to take some or all of their course-based exams through the services of the DRC's accommodated testing unit. Still other students may seek out the services of the Purdue Testing Center to take a professional certification, equivalency, or graduate entrance exam (among others). These are all separate units with differing reporting structures and very small staffs.

The existing facilities are not purpose-built. Therefore, each unit in isolation is limited in its ability to serve students. For instance, a two-person unit that is short one person cannot deliver a test (in some cases) due to the demand for two proctors. By bringing these units together in one unit and one space, the depth in staff and the purpose-built facility will be able to expand the ability of the comprehensive unit to serve more students in more ways.

In addition to the benefits to ease of use, there's an overall benefit to campus as well. By combining several small testing centers into one large robust center, an efficiency of scale can be realized, and more students can be serviced with less area. The net increase in space allocated for student testing is 4,145 square feet. However, the proposed testing center can service 247 percent more students in 44 percent less area per student than the existing, individual testing centers.

Overview of Combined Units

The primary accommodation offered by the Disability Resource Center is accommodated testing. Because the DRC will be a major contributor of students to the central testing center, there is a strong relationship between the DRC and the proposed CTC. During mid-term and final exam periods, the demand for accommodated testing exceeds the capacity of the DRC's testing space by a factor

greater than 10. The DRC is then forced to use libraries and other spaces to house accommodated exams. Additionally, the DRC staff must be augmented by volunteer and paid proctors from its parent unit (Student Success) and other units at Purdue. This Central Testing Center will create the ability for accommodated testing to occur in a much more efficient and controlled way and with a deeper overall testing staff (to oversee the inevitable volunteer proctors).

Foreign Language Testing is currently conducted in a general use computer laboratory. During summer orientation (STAR) this space becomes overcrowded, loud, and inefficient. This department would benefit greatly from a purpose built, high capacity testing center.

The Digital Education Testing Center is an on-campus proctored testing site that administers a variety of paper-based and computerized exams. Tests administered in this center are for Purdue-based distance learning courses, as well as CASA (Core Academic Skills Assessment) and Pearson Vue exams.

The Purdue Testing Center serves campus and patrons outside of campus by providing a variety of testing services. They are the only center on campus fully certified by national testing organizations and offer national registration and licensing exams. The tests administered by this department in addition to some tests given by the Digital Education Testing Center, represent the core of tests that will be administered in the high stakes testing area of the Central Testing Center.

Existing Testing Centers	Seats	Area	Tests/Day*	SF/Seat
Disability Resource Center, Accommodated Testing	24	1,515	15	63
Purdue Testing Center	16	952	30	60
Digital Education	25	2,790	20	112
Foreign Language Placement*	36	1,744	30	48
Total	**101**	**7,001**	**95**	**283**
Average	**25**	**1,750**	**24**	**71**

* Foreign Language Placement testing does not occur year round. Area shown does not include administration space.

Proposed Central Testing Center	Seats	Area	Tests/Day*	SF/Seat
Central Testing Center - Option E and F	250	11,416	TBD	46
Average	**250**	**11,416**		**46**

3. COLLEGE OF ENGINEERING

In 2014, the College of Engineering completed a campuswide master plan and identified the following goals.

- Provide appropriate office and office support space for the anticipated number of personnel
- Create a state-of-the-art teaching and learning environment
- Create a state-of-the-art research environment
- Incorporate state-of-the-art technology
- Create collaboration and interaction environments
- Enhance Purdue's close relationship with state and industry partners
- Create a community with a sense of place that proclaims the stature of the College of Engineering
- Align with the Campus Master Plan

This plan was driven by the need to provide more and greater quality space to meets the College's strategic growth initiative, including increased student enrollment and 30 percent personnel growth, as well as the need to efficiently and strategically accommodate that growth. Within that plan, the A.A. Potter Center was identified as the ideal home to co-locate the COE's programs functions along with First-Year Engineering, since both of these programs focus on outreach toward, and engagement of, potential engineers. The Division of Environmental and Ecological Engineering was also identified as ideal for location in Potter due to its need to have a discrete home separate from Civil Engineering.

College of Engineering Programs

The Programs of the College of Engineering consist of Women in Engineering Program, Minority Engineering Program, the Office of Professional Practice, the Global Engineering Program, Honors, and the Office of Future Engineers. These programs share commonalities in the populations they serve and, in some cases, the types of facilities that are needed. Locating these programs together will develop synergies and generate efficiencies of space utilization that would not be possible if they were dispersed across the campus.

Engineering Education

Engineering Education has a critical need for additional space immediately. ENE is expected to grow over 200 percent in the future and will ultimately require over 70,000 nsf. ENE has an educational mission of serving all first-year engineering students and requires large and flexible instructional spaces. ENE's focus on first-year engineering students creates a strong synergy with the COE's Programs Functions.

The Division of Environmental and Ecological Engineering

The Division of Environmental and Ecological Engineering is expected to grow and become an independent School of the College of Engineering. The projected growth in space required for EEE is over 600 percent. Identifying a permanent home that can satisfy EEE needs for space and contribute to its independent identity was a major focus of the COE's master planning effort. Potter was identified as an ideal location due to the synergy between EEE and the School of Industrial Engineering in Grissom Hall.

4. FOOD SERVICE

The following food service options can be found within a two-minute walk of the A.A. Potter Center.

- Beans Café, featuring coffee and coffee drinks, deli items, baked goods, snacks, and soda
- Catalyst Café, serving smoothies, coffee and coffee drinks, baked goods, grab-and-go sandwiches and salads
- Au Bon Pain Restaurant, serving baked goods, coffee, coffee drinks, tea, custom or pre-made sandwiches and salads, soups, breakfast sandwiches, pastries, wraps, fruit, and snacks.

Note that all the options above are in separate locations; thus, a student is only presented with a single dining option at each location. Au Bon Pain offers the most menu options but is still limited in food style. Also worth noting is the popularity of Au Bon Pain. The extensive amount of student gathering/study/hangout spaces within the Wilmeth Active Learning Center has created a large concentration of potential customers for the restaurant. The proposed inclusion of food service within the A.A. Potter Center would expand the possible dining options from one to four restaurants immediately accessible to these students.

Also of note, the proposed redesign of the exterior plaza dining space on the south side of Potter will integrate the building and plaza into the centennial mall project. This will create outdoor dining opportunities that aren't present elsewhere in this area.

CONSTRUCTION COST SUMMARY

	GSF	COST PER SFT	TOTAL CONSTRUCTION COST
OPTION D	waiting	waiting	waiting
OPTION E	waiting	waiting	waiting

CAMPUS SPACE USE IMPACT ANALYSIS

The chart on the following page assesses the impact of each proposed option on space utilization within the A.A. Potter Center and Purdue campus as a whole.

All areas affected are categorized using the FCIM Classification system and are accounted for in this chart. It should be noted that the “310 – Office” column shows space allocated to that specific category for informational purposes. The area shown in this column is already contained in the “300 – Offices” category; therefore, this column is not included in calculations.

OPTION E

The impact of eliminating spaces from Option D to open up areas for dining can be seen here. The largest decrease relative to Option D comes in instructional lab space, resulting from the removal of several large classrooms from the project. The impact of this shift can be seen in the increase in general use space, which accounts for the addition of the dining functions to building. A slight increase can be seen in the 300 office category. This primarily results from the reclassification of a large percentage of the first floor corridor space, which had been classified as general circulation space; but in Option E, where it is enclosed and behind a reception area, this corridor has now been classified as office service space.

OPTION F

In terms of space usage, Option E and F are nearly identical. The additional square footage created by the new elevator and mechanical shaft outside the building can be seen in the 1,000-sft increase in net area. The new floor area ends up distributed somewhat evenly across several categories without remarkable impacts.

CAMPUS SPACE USE IMPACT ANALYSIS

		100 - Classroom	200 - Instructional Labs	250 - Research Labs	300 - Offices	* 310- Offices	400 - Study	500 - Special Use	600 - General Use	700 - Support	800 - Health Care	900 - Residential	050 - Unclassified	000 - General Building	TOTAL NSF	TOTAL ASF
Option E	Space Renovated/Created in Potter	0	6,260	0	42,066	33,164	11,014	0	16,262	0	0	0	0	28,563	104,165	75,602
	Space Displaced from Potter	3,115	4,690	28,736	19,748	13,641	18,182	202	732	1,947	0	0	0	16,514	93,866	77,352
	Net Change in Potter Total = (Renovated) - (Displaced)	**-3,115**	**1,570**	**-28,736**	**22,318**	**19,523**	**-7,168**	**-202**	**15,530**	**-1,947**	**0**	**0**	**0**	**12,049**	**10,299**	**-1,750**
	Space Released on Campus	0	7,587	469	25,376	19,891	3,792	0	21	1,624	0	0	0	150	39,019	38,869
	Space Displaced from Potter	-3,115	-4,690	-28,736	-19,748	-13,641	-18,182	-202	-732	-1,947	0	0	0	-16,514	-93,866	-77,352
	Space Accommodated Elsewhere	3,115	0	0	2,621	1,414	17,629	202	0	0	0	0	0	775	24,342	23,567
	Net Change on Campus Total = (Released) + (Displaced) + (Accommodated)	**0**	**2,897**	**-28,267**	**8,249**	**7,664**	**3,239**	**0**	**-711**	**-323**	**0**	**0**	**0**	**-15,589**	**-30,505**	**-14,916**

		100 - Classroom	200 - Instructional Labs	250 - Research Labs	300 - Offices	* 310- Offices	400 - Study	500 - Special Use	600 - General Use	700 - Support	800 - Health Care	900 - Residential	050 - Unclassified	000 - General Building	TOTAL NSF	TOTAL ASF
Option F	Space Renovated/Created in Potter	0	5,121	0	42,882	33,152	11,088	0	17,089	0	0	0	0	28,935	105,115	76,180
	Space Displaced from Potter	3,115	4,690	28,736	19,748	13,641	18,182	202	732	1,947	0	0	0	16,514	93,866	77,352
	Net Change in Potter Total = (Renovated) - (Displaced)	**-3,115**	**431**	**-28,736**	**23,134**	**19,511**	**-7,094**	**-202**	**16,357**	**-1,947**	**0**	**0**	**0**	**12,421**	**11,249**	**-1,172**
	Space Released on Campus	0	7,587	469	25,376	19,891	3,792	0	21	1,624	0	0	0	150	39,019	38,869
	Space Displaced from Potter	-3,115	-4,690	-28,736	-19,748	-13,641	-18,182	-202	-732	-1,947	0	0	0	-16,514	-93,866	-77,352
	Space Accommodated Elsewhere	3,115	0	0	2,621	1,414	17,629	202	0	0	0	0	0	775	24,342	23,567
	Net Change on Campus Total = (Released) + (Displaced) + (Accommodated)	**0**	**2,897**	**-28,267**	**8,249**	**7,664**	**3,239**	**0**	**-711**	**-323**	**0**	**0**	**0**	**-15,589**	**-30,505**	**-14,916**

* This value is also contained in "300 - Offices"

EFFICIENCY ANALYSIS PER DEPARTMENT

	Option	% Project	ASF	Faculty/Staff	Grad	Office ASF	Total Headcount	FTE	Office ASF/ Headcount	Office ASF/FTE
Disability Resource Center	Existing		3,427	14		1,626	14	14.0	116	116
	E	7.74%	3,356	15	2	1,802	17	15.9	106	114
	F	7.74%	3,358	15	2	1,793	17	15.9	105	113
Assistive Technology Center	Existing		998	1		150	1	1.0	150	150
	E	2.24%	971	1		138	1	1.0	138	138
	F	2.24%	971	1		138	1	1.0	138	138
Central Testing Center [1] (all departments)	Existing		6,564	13		2,851	13	13.0	219	219
	E	26.18%	11,346	18		1,058	18	18.0	59	59
	F	26.43%	11,471	18		1,058	18	18.0	59	59
Instructional Data Processing	Existing		2,230	5		534	5	5.0	107	107
	E	2.55%	1,106	6		721	6	6.0	120	120
	F	2.55%	1,106	6		721	6	6.0	120	120
Online Compliance [2]	Existing									
	E	1.15%	500	8		500	8	8.0	63	63
	F	1.15%	500	8		500	8	8.0	63	63
Business Office [2]	Existing									
	E	1.45%	630	7		630	7	7.0	90	90
	F	1.45%	630	7		630	7	7.0	90	90
Engineering Education	Existing		16,845	77		8,022	77	77.0	104	104
	E	24.86%	10,772	46	146	10,563	192	109.9	55	96
	F	24.82%	10,770	46	146	10,561	192	109.9	55	96
Minority in Engineering	Existing		1,525	7		680	7	7.0	97	97
	E	3.54%	1,533	6		567	6	6.0	95	95
	F	3.53%	1,533	6		567	6	6.0	95	95
Office of Future Engineers	Existing		694	9		694	9	9.0	77	77
	E	1.38%	596	4	3	596	7	5.3	85	112
	F	1.37%	596	4	3	596	7	5.3	85	112

	Option	% Project	ASF	Faculty/Staff	Grad	Office ASF	Total Headcount	FTE	Office ASF/ Headcount	Office ASF/FTE
Women in Engineering	Existing		1,072	13		849	13	13.0	65	65
	E	3.59%	1,555	10	11	1,555	21	14.8	74	105
	F	3.58%	1,555	10	11	1,555	21	14.8	74	105
Engineering Honors	Existing		829	4		474	4	4.0	119	119
	E	4.94%	2,140	10		869	10	10.0	87	87
	F	4.93%	2,139	10		869	10	10.0	87	87
Environmental & Ecological Engineering	Existing		3,738	25		2,412	25	25.0	96	96
	E	18.76%	8,130	34	54	4,916	88	57.6	56	85
	F	18.60%	8,074	34	54	4,916	88	57.6	56	85
Office of Professional Practice	Existing		1,592	8		1,016	8	8.0	127	127
	E	1.61%	697	9		697	9	9.0	77	77
	F	1.61%	697	9		697	9	9.0	77	77
Dining [2] Food Service	Existing									
	E	P:\2015\02663\F.	3,550	0	0	0	0	0.0	0	0
	F	8.17%	3,546	0	0	0	0	0.0	0	0

[1] This department is the combination of several existing departments and is offering an expanded program in the proposed design.
[2] New Department

AREA ANALYSIS PER DEPARTMENT

	Option	Class-rooms	Inst. Lab	Res. Lab	Offices	Study	Special Use	General Use	Support	Health Care	Residen-tial	General Building	Total ASF	ASF % Change
Disability Resource Center	Existing	0	0	0	2,349	1,078	0	0	0	0	0		3,427	
	E	0	0	0	3,358	0	0	0	0	0	0	0	3,358	-2%
	F	0	0	0	3,358	0	0	0	0	0	0	0	3,358	-2%
Assistive Technology Center	Existing	0	0	0	150	848	0	0	0	0	0		998	
	E	0	0	0	138	833	0	0	0	0	0	0	971	-3%
	F	0	0	0	138	833	0	0	0	0	0	0	971	-3%
Central Testing Center [1] (all departments)	Existing	0	1,744	0	2,946	1,874	0	0	0	0	0		6,564	
	E	0	0	0	1,519	9,215	0	0	0	0	0	612	10,734	64%
	F	0	0	0	1,570	9,289	0	0	0	0	0	612	10,859	65%
Instructional Data Processing	Existing	0	0	0	606	0	0	0	1,624	0	0		2,230	
	E	0	0	0	1,106	0	0	0	0	0	0	0	1,106	-50%
	F	0	0	0	1,106	0	0	0	0	0	0	0	1,106	-50%
Online Compliance [2]	Existing													N/A
	E	0	0	0	500	0	0	0	0	0	0	0	500	
	F	0	0	0	500	0	0	0	0	0	0	0	500	
Business Office [2]	Existing													N/A
	E	0	0	0	630	0	0	0	0	0	0	0	630	
	F	0	0	0	630	0	0	0	0	0	0	0	630	
Engineering Education	Existing	0	5,843	0	10,507	0	0	21	0	0	0		16,371	
	E	0	0	0	10,772	0	0	0	0	0	0	0	10,772	-34%
	F	0	0	0	10,770	0	0	0	0	0	0	0	10,770	-34%
Minority in Engineering	Existing	0	0	0	1,008	517	0	0	0	0	0		1,525	
	E	0	0	0	567	966	0	0	0	0	0	0	1,533	1%
	F	0	0	0	567	966	0	0	0	0	0	0	1,533	1%
Office of Future Engineers	Existing	0	0	0	694	0	0	0	0	0	0		694	
	E	0	0	0	596	0	0	0	0	0	0	0	596	-14%
	F	0	0	0	596	0	0	0	0	0	0	0	596	-14%

	Option	Class-rooms	Inst. Lab	Res. Lab	Offices	Study	Special Use	General Use	Support	Health Care	Residen-tial	General Building	Total ASF	ASF % Change
Women in Engineering	Existing	0	0	0	1,072	0	0	0	0	0	0		1,072	
	E	0	0	0	1,555	0	0	0	0	0	0	0	1,555	45%
	F	0	0	0	1,555	0	0	0	0	0	0	0	1,555	45%
Engineering Honors	Existing	0	0	0	829	0	0	0	0	0	0		829	
	E	0	1,271	0	869	0	0	0	0	0	0	0	2,140	158%
	F	0	1,270	0	869	0	0	0	0	0	0	0	2,139	158%
Environmental & Ecological Engineering	Existing	0	0	469	2,716	553	0	0	0	0	0		3,738	
	E	0	2,836	0	5,294	0	0	0	0	0	0	0	8,130	117%
	F	0	2,838	0	5,236	0	0	0	0	0	0	0	8,074	116%
Office of Professional Practice	Existing	0	0	0	1,592	0	0	0	0	0	0		1,592	
	E	0	0	0	697	0	0	0	0	0	0	0	697	-56%
	F	0	0	0	697	0	0	0	0	0	0	0	697	-56%
Dining [2] Food Service	Existing													
	E													
	F													

[1] This department is the combination of several existing departments and is offering an expanded program in the proposed design.
[2] New Department

POTTER CENTER RENOVATION IN RELATION TO THE "PURDUE MOVES" CAMPUS INITIATIVE

> “This is no ordinary time for higher education, and we are determined to seize the opportunity to increase our investment in areas that have the greatest potential for impact. Our students will grow, not just academically, but in ways that truly prepare them to effectively work and live in today's world. Our research will deliver still greater benefits to our state and to all corners of the globe. And we will achieve this in a way that is fiscally responsible.”
>
> – Mitch Daniels

Any major project undertaken on any campus must be examined in light of the overall goals and direction for that campus. Here we will examine the A.A. Potter Center Renovation through the lens of the “Purdue Moves” campus initiative.

Initiatives	Strongly Support	Support	No Impact
Affordability and Accessibility			
Increasing Value	●		
Expanding Opportunity		●	
STEM Leadership			
Expanding Engineering	●		
Transforming Technology			●
Strengthening Computer Science			●
World-Changing Research			
Drug Discovery			●
Plant Sciences Research			●
Research Communication			●
Transformative Education			
Transforming Teaching and Learning	●		
International Experiences	●		
Living and Learning		●	
Year-Round University		●	

AFFORDABILITY AND ACCESSIBILITY

Increasing Value
By renovating an existing building rather than constructing a new building, Purdue will substantially reduce construction costs. Additionally, more efficient space utilization in the proposed plans reduces the demand for additional area on campus. Renovations that increase spatial efficiency also support a key principal of the campus' master plan—the promotion of compact growth.

By creating a larger, combined central testing center and capitalizing on the economy of scale, the University will realize large efficiencies of space and staff within these departments. This will release space on campus for other uses and decrease the need for the University to construct new space. This effect will be multiplied during final exam periods by greatly reducing or eliminating the need to create improvised testing centers across campus. These impromptu centers result in the inefficient use of a large quantity of space and resources.

Expanding Opportunity
The expansion and relocation of the Disability Resource Center will create a greater ease of use and access to students with disabilities, allowing more students to successfully attend and navigate the university and focus on success.

STEM LEADERSHIP

Expanding Engineering
The College of Engineering's programs/departments focus on the recruitment and retention of students. Co-location and expansion of these programs will result in greater efficiency and effectiveness, which will directly correlate with Purdue's goals of increasing engineering graduate student enrollment and ultimately increase the number of graduating engineers.

Construction of a 21st century learning environment will attract, recruit, and retain students and faculty. This will directly contribute to the success and growth of the College of Engineering.

WORLD-CHANGING RESEARCH

Drug Discovery, Plant Sciences Research, and Research Commercialization
Because the full renovation option proposes to displace research lab space for classroom and office space, the Potter renovation may be at odds with this goal. In this instance, the goal of expanding engineering directly affects the goal of investing in research. It should be noted that it is important to consider the age and condition of the research lab space in Potter when assessing its true value to campus. A full assessment of the existing space is attached in the appendix to this study.

TRANSFORMATIVE EDUCATION

Transforming Teaching and Learning
The College of Engineering strongly believes in the i2i or Ideas in Innovation style educational model. Additional i2i spaces will be created in the renovated Potter Center. Additionally, the Potter Center will be home to a multitude of flexible non-traditional study spaces—study lounges, meeting rooms, conference rooms, work cafés, and tutorial centers—supporting a 21st century educational model.

International Experiences
A critical driver to improving the number of students studying aboard is increasing student awareness of these opportunities. This is the primary focus of the College of Engineering's Global Engineering Program. This program is one of many of the COE's programs proposed for location on the ground floor of Potter. This location will create a synergy between the GEP and all the COE's programs focused around the common goal of student success and advising. This synergy is not currently realized.

Living and Learning
The proposed Potter Center renovation does not contain a student housing component, so it cannot be a direct contributor to the integration of learning space into residential space. However, the center is located in a key location along the student success corridor. Therefore, its utilization could play a major role in the success of the corridor, which is a major contributor to the Living and Learning initiative. The uses proposed for the building appear to be ideally suited to integrate with this corridor as they focus on the retention and attraction of students, access to testing, ability for disabled students to interface with the University, and general study and educational spaces. Additionally the proposed redesign of the Potter south terrace creates an outdoor classroom and gathering space, a function not currently found within the corridor.

Year-Round University
By concentrating the College of Engineering's Programs in one location, students will have enhanced access to advisors, who can help guide students toward summer programs, shortening student's time to graduation.

Summer online courses have been identified as a focus area for the expansion of summer education. The creation of the central testing center will greatly support online courses by creating an efficient, effective, and academically rigorous testing environment. It will also expand students options for testing by creating a greater ease of testing access.

PURPOSE AND OBJECTIVES

American Structurepoint, in conjunction with Applied Engineering and Blundall Associates, has been invited to conduct the following study regarding the future use of the A.A. Potter Engineering Center.

This study will holistically examine the building's potential to accommodate seven different functions:

- The Disability Resource Center (DRC)
- The Assistive Technology Center (ATC)
- The New Online Compliance Department
- A new campus Centralized Testing Center (CTC)
- The Instructional Data Processing (IDP) Department
- Relocation of nine College of Engineering (COE) departments in accordance with their overall master plan
- Multiple food service vendors with indoor and outdoor dining areas

GOALS

The intent of this study is to propose a renovation that accomplishes four distinct goals. Note that not all of these goals can be accomplished with most options presented here. Only Option D fully meets these aims:

- Create a more accessible and user-friendly DRC and increase efficiency and coordination between the DRC and related departments through co-location of those departments.
- Increase efficiency, cost control, academic integrity, testing flexibility, and ease of use for students and staff by combining all the campus testing functions into one location to serve all of campus.
- Realize the expansion and growth goals of the College of Engineering by strategically relocating departments identified by the College's master plan in accordance with their Strategic Initiatives Plan.
- Successfully integrate food service into Potter Center and use the draw of food service to expose students to the field of engineering.

It is important to note that the building is currently occupied by several departments of the College of Engineering (COE). The COE recently completed a holistic master plan to study the current and future spatial needs of the college in order to accomplish the goals of the COE's Strategic Plan. Much of the space within Potter Center was allocated for COE use as part of that master plan. The new departments being proposed for inclusion in Potter as part of this study were not accounted for in the COE master plan. It is the overarching goal of this study to propose a design solution that simultaneously respects the needs of and supports the future goals for the College of Engineering while co-locating the proposed functions within the building and capitalizing on inter-departmental synchronicities where possible.

This study will examine the work associated with the proposed renovations/relocations in five separate renovation scenarios outlined below, each with sequentially increasing scope and budget.

Option E

- Relocation of the DRC and Assistive Technology Center
- Creation of a Centralized Testing Center
- Relocation of the Instructional Data Processing Department

- Renovation of the entirety of the remaining building to accommodate the College of Engineering's Master Plan
- Creation of a new building entrance and selective "reskinning" of the façade
- Selective revision of the currently proposed plan for Centennial Mall to holistically integrate the Potter Center into campus
- Creation of multiple retail dining venues on the first floor of the Potter Center serving an indoor and outdoor dining area

Option F

- Relocation of the DRC and Assistive Technology Center
- Creation of a Centralized Testing Center
- Relocation of the Instructional Data Processing Department
- Renovation of the entirety of the remaining building to accommodate the College of Engineering's Master Plan
- Creation of a new building entrance and selective "reskinning" of the façade
- Selective revision of the currently proposed plan for Centennial Mall to holistically integrate the Potter Center into campus
- Creation of multiple retail dining venues on the first floor of the Potter Center serving an indoor and outdoor dining area
- Expansion of the building footprint to accommodate an elevator and new mechanical shaft to serve all four floors of Potter, creating open access and views across the dining areas

PROJECT PARTICIPANTS

PURDUE UNIVERSITY CONTRIBUTORS

The design team would like to thank the following Purdue University Faculty and Staff for their time and contribution to this study.

Donna Ahlen – Manager, Facilities Planning and Construction
Michele Arthur – Assistant Director Equity in Hiring
Jason Baumer – Continuing Lecturer in German; Course Coordinator
Dean Brusnighan – Assistive Technology Specialist Consulting and Training
Brad Bowen – University Engineer and Senior Director of Asset Management
Dan Carpenter – Director of Student Success at Purdue
Frank Dooley – Vice Provost for Teaching and Learning
Michael Eddy – Senior Associate Director, Digital Education
Mary Ann Ferkis – Assistant Director, Disability Resource Center
Cindy Fields – Purdue Testing Center Manager
Kevin Flanagan – Digital Education
Robert Frosch – Associate Dean for Resource Planning and Management, Professor of Civil Engineering
Caren Hanley – Disability Resource Center
Madeleine Henry – Head, School of Languages and Cultures
Aaron Hunley – Capitol Project Manager
Phil Knobloch – Audio Visual Services Manager
Maricel Lawrence – Associate Director for Digital Education
Mark Lehman – Capitol Project Manager
Laura Miller – Instructional Data Processing
David Nelson – Associate Director of the Center for Instructional Excellence
Kim Rechkemmer – Assistant Director for Space Management
Byron Reed – ITaP Director of Learning Spaces
David Schwarte – Assistive Technology Specialist Consulting and Training
Randall Ward – Disability Resource Center Director

DESIGN TEAM

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Blundall Associates | COST ESTIMATING
Martyn Blundall
Phillip Salisbury
Bryan Smedberg

Reitano Design Group | FOOD CONSULTING
Scott Reitano
Chris Wair

BUILDING HISTORY AND CURRENT USE

The A.A. Potter Engineering Center (POTR) is a multi-purpose teaching and research facility administered by the Institute for Interdisciplinary Engineering Studies. The Center's mission is to develop and promote innovative technologies in response to society's needs. Its goal is to enhance quality of life.

The Center is named for Andrey Abraham Potter, the dean of engineering at Purdue from 1920 to 1953. Dean Potter built Purdue into the largest and one of the most respected engineering colleges in the country. From 1945 to 1946, he served as acting president of Purdue; he was offered the presidency but declined.

The Potter building is centrally located on the north campus, immediately to the east of the Wilmeth Active Learning Center (currently under construction). The building currently houses the Siegesmund Engineering Library, the Office of Professional Practice, Visual Analytics for Command, Control, and Interoperability Environments (VACCINE), Laboratory of Renewable Resources Engineering (LORRE), VLSI Design and Test Laboratory, the Center for Materials Under eXtreme Environment (CMUXE), the offices of the Division of Environmental and Ecological Engineering, and the CSOPS Research Entity for School Research.

The building houses two named spaces. It is critical the following spaces be accommodated for and named in the newly renovated space.

K.S. FU ROOM FOR CROSS-DISCIPLINARY RESEARCH

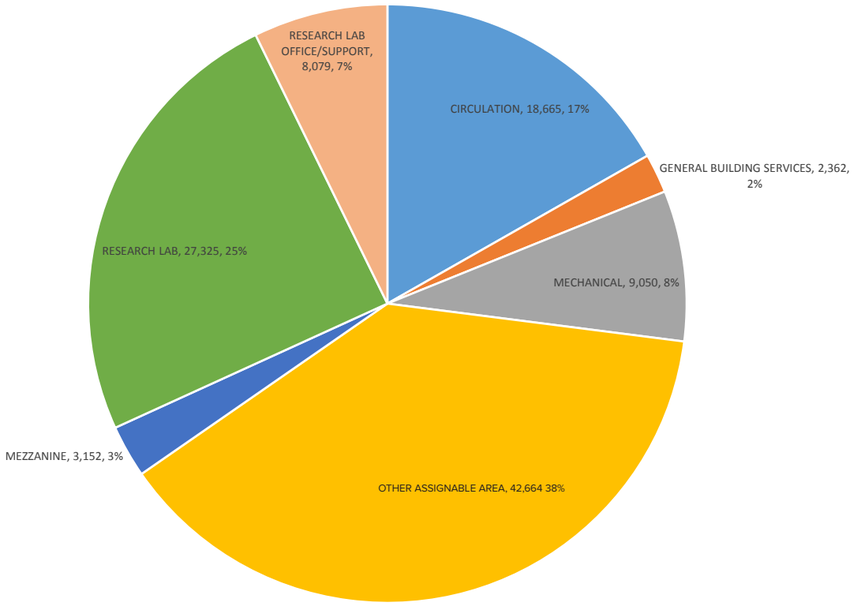
The K.S. Fu Room for Cross-Disciplinary Research (Room 234) is a meeting room that comfortably seats up to 50 people and has been equipped for cross-disciplinary work, special research forums, site visits, cross-disciplinary seminars, presentations, workshops, roundtables, and the display of publications, posters, and audio/visual materials.

This conference room was named in recognition of the late Professor King-Sun Fu because of his many accomplishments as Goss Distinguished Professor of Engineering. Professor Fu was a member of the faculty of the Purdue School of Electrical Engineering from 1960 to 1985.

THE MALOTT COMMONS

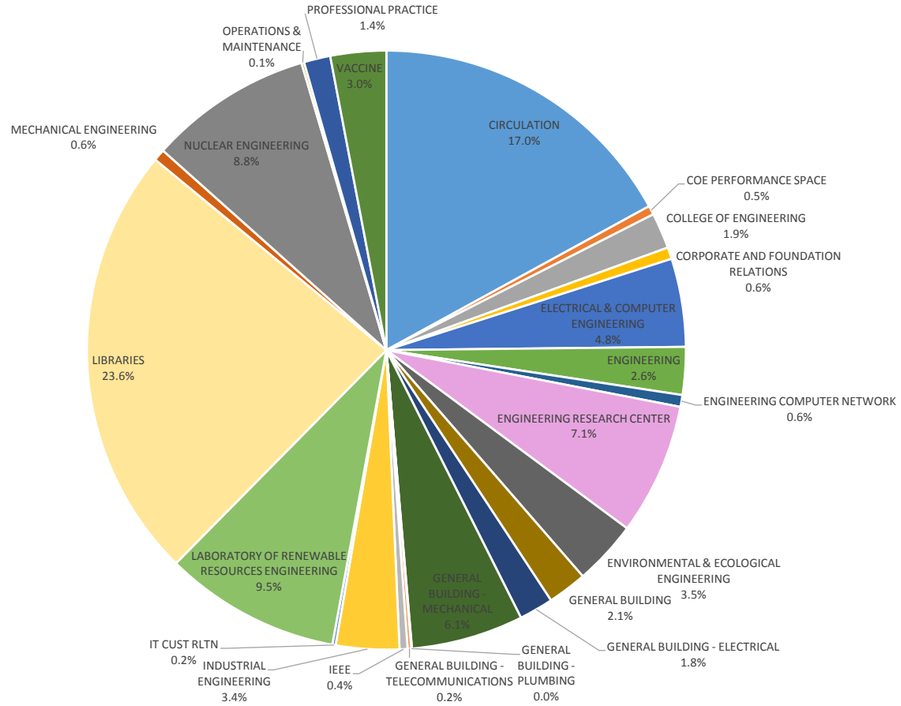
This common area honors Thomas J. Malott (BSME 1962, HDR 2002) and Sandra H. Malott, who donated in support of the Office of Professional Practice. The commons is an entity shared by the entire Potter Center.

POTTER CENTER EXISTING USE



Mezzanine – Area excluded from all options
Research Labs – Area excluded from options A, B, and C
Research Lab Office/Support – Area excluded from options A, B, and C
Mechanical – Area excluded from all options
Other Assignable Areas – Varying amount included in all options
Circulation – Varying amount included in all options

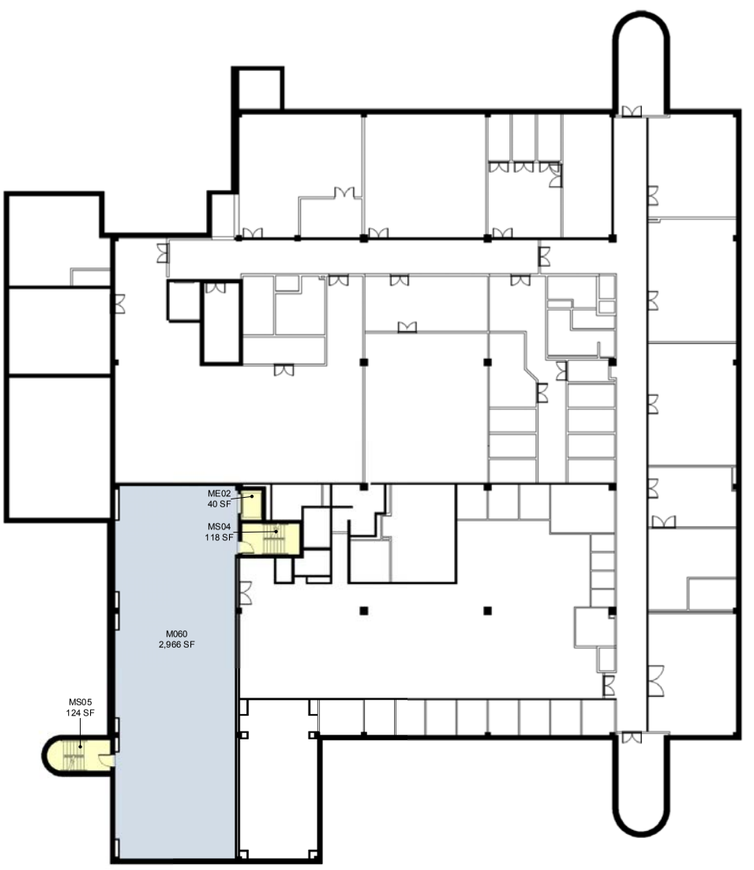
POTTER CENTER EXISTING DEPARTMENT USE





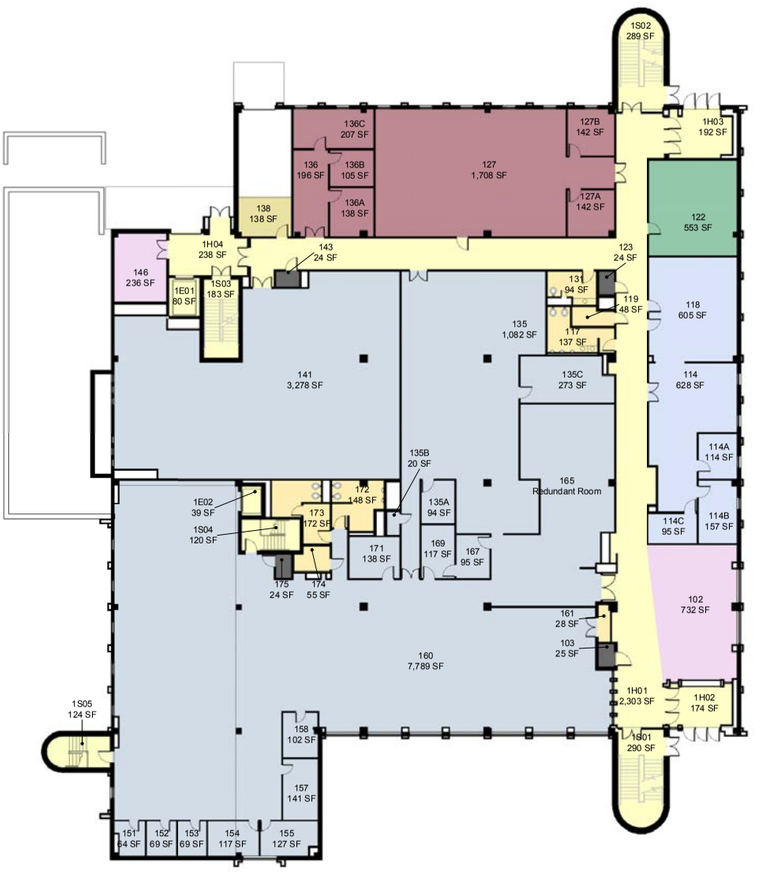
- CIRCULATION
- COE FLEX SPACE
- COLLEGE OF ENGINEERING
- CORPORATE AND FOUNDATION RELATIONS
- ELECTRICAL & COMPUTER ENGINEERING
- ENGINEERING COMPUTER NETWORK
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENGINEERING RESEARCH CENTER
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- IEEE
- INDUSTRIAL ENGINEERING
- IT CUST RLTN
- LABORATORY OF RENEWABLE RESOURCES ENGINEERING
- LIBRARIES
- MECHANICAL ENGINEERING
- NUCLEAR ENGINEERING
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- VACCINE

EXISTING LOWER LEVEL

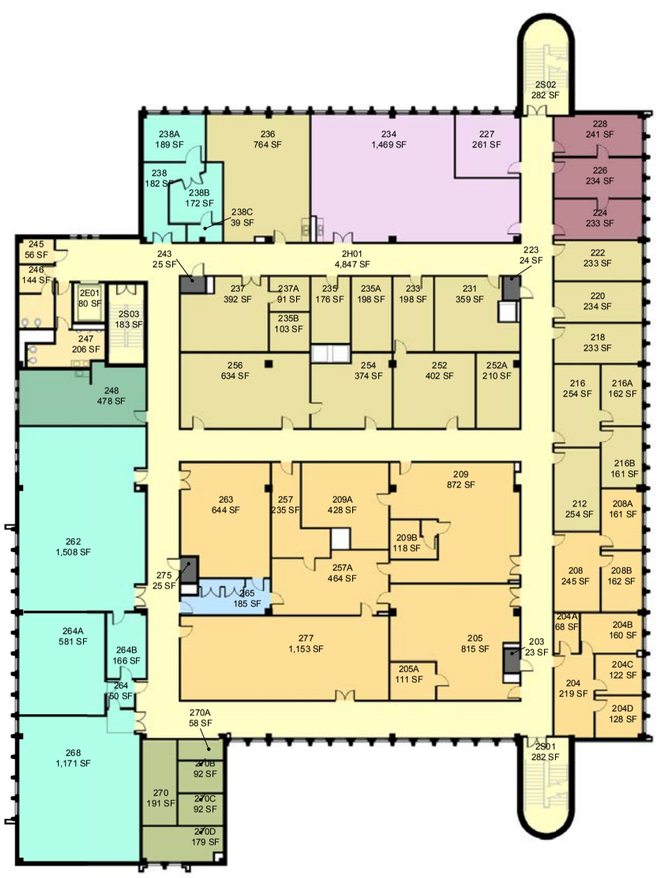


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- VACCINE

EXISTING MEZZANINE



EXISTING FIRST FLOOR

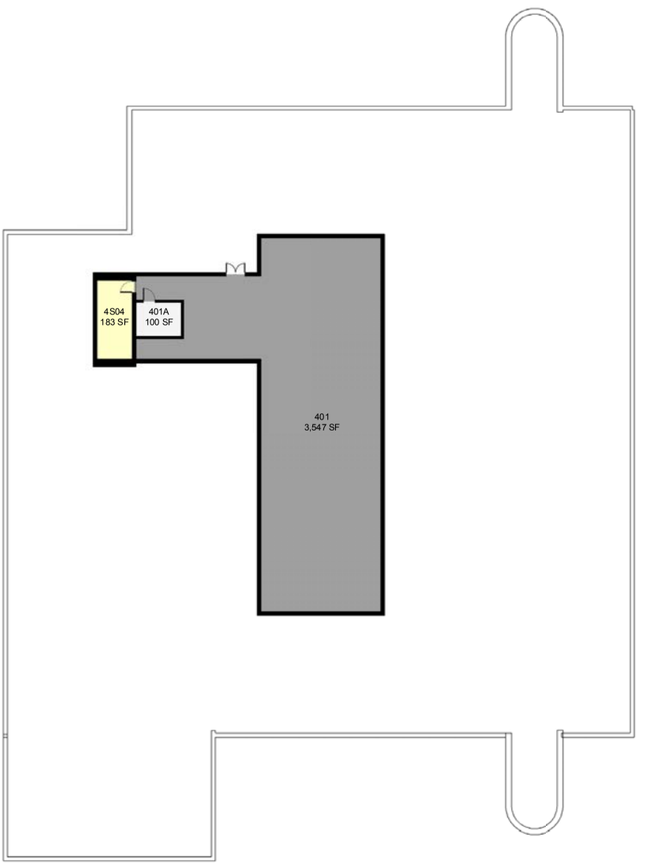


EXISTING SECOND FLOOR



- CIRCULATION
- COE FLEX SPACE
- COLLEGE OF ENGINEERING
- CORPORATE AND FOUNDATION RELATIONS
- ELECTRICAL & COMPUTER ENGINEERING
- ENGINEERING COMPUTER NETWORK
- ENGINEERING EDUCATION
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- PROFESSIONAL PRACTICE
- RESTROOM
- VACCINE

EXISTING THIRD FLOOR



- CIRCULATION
- COE FLEX SPACE
- COLLEGE OF ENGINEERING
- CORPORATE AND FOUNDATION RELATIONS
- ELECTRICAL & COMPUTER ENGINEERING
- ENGINEERING COMPUTER NETWORK
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- VACCINE

EXISTING PENTHOUSE

PHYSICAL ANALYSIS

Date of Construction.....1974

Number of Stories.....3 above grade
1 below grade
1 partial mezzanine
1 penthouse

Area .....Basement: 30,629 sft
Mezzanine: 3,712 sft
First Floor: 28,003 sft
Second Floor: 27,952 sft
Third Floor: 27,776 sft
Penthouse: 4,236 sft
TOTAL AREA: 122,308 sft
TOTAL OCCUPIABLE AREA: 114,360 sft

Sprinkler System .....Fully sprinklered

Floor-to-Floor Heights .....Basement to First: 15’-4”
Basement to Mezzanine: 7’-8”
Mezzanine to First: 7’-8”
First to Second: 14’-0”
Second to Third: 14’-0”
Third to Roof: 14’-0”

Cladding.....Exterior

Typical Brick Cavity Wall Veneer Assembly.....3 5/8” brick
1 7/8” airspace
1/2” impregnated sheathing board
6” 20-gauge metal stud at 16”
on center with 6” batt insulation
Vapor barrier
1/2” gypsum board

Typical Precast Concrete Panel Exterior Wall Assembly .....6” to 8” precast concrete panel
(thickness varies)
1/2” Airspace
1/2” impregnated sheathing board
6” 20-gauge metal stud at 16”
on center with 6” batt insulation
Vapor barrier
1/2” gypsum board

R Value: .....11.14 (wall construction does not meet
current code requirements
for insulation)

Current Replacement Value $54,756,601*

Deferred Maintenance Value $7,930,500*

*as reported according to the 2014 study produced by The SLAM Collaborative

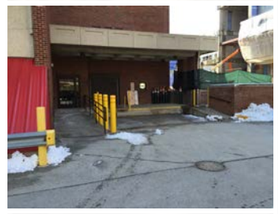
ACCESSIBILITY

Though examination of accessibility to disabled persons is always important, it is especially important to this study because of the proposed relocation of the Disability Resource Center to the Potter Center.

Because the DRC serves Purdue’s disabled student population, this building will be among the most heavily trafficked buildings on campus by students with special needs. While the entire building should be remodeled to be ADA compliant, areas specifically serving the DRC should be designed above and beyond code requirements to accommodate these students to the greatest extent possible.

As outlined in the following analysis, in its current condition, the Potter Center is not an accessible building. Therefore, it is the opinion of this study that the Center would not be fit for use by the DRC unless the DRC’s relocation occurred concurrently with accessibility upgrades.









PARKING

The Potter Center is not served by a designated parking lot. However, there are three parallel parking spaces along the east side of the alley immediately east of Potter. These spaces are not in an ideal location close to an existing ramp to access the building; restriping and labeling these as ADA spaces appears to be the best available solution. Because the building is bound to the west and south by pedestrian-only roads, there is limited opportunity to create alternate parking solutions. A sidewalk should be constructed connecting these spaces to the sidewalks and entrance at the southeast entrance. There is currently no sidewalk along this side of the building.

BUILDING ENTRANCES AND ENTRANCE ACCESS

The finished floor level of the Potter Center is approximately 2' above grade level; therefore, the only accessible routes to the building are by ramp. Three ramps are present—one each at the northwest, northeast, and southeast corners. The ramp at the northwest corner is adjacent to the building's loading dock and is primarily intended to function as a service ramp. The door at the end of this ramp leads toward the freight elevator, which is currently the building's only elevator.

The ramp near the southeast corner entrance is 106' from the southeast door. It is somewhat hidden behind a wall along the far west edge of the exterior terrace. A student in a wheelchair wishing to use the southwest entrance (the building's main entrance) would need to travel between 138 and 274' farther than an able-bodied peer. Because the south side of the building is along a pedestrian only path, the ramp itself is easily accessible.

ELEVATORS

The building currently contains a single freight elevator at the northwest corner of the building. Aesthetically, the freight elevator is not welcoming and is not a pleasure to use. On an emotional level, relegating wheelchair-bound students to ride with the building's cargo is not hospitable.

To examine a worst-case scenario, starting at a point at the southeast corner outside the building, an able-bodied student wishing to use the building's main entrance and travel to room 205 at the southeast corner of the second floor would travel 186'. A wheelchair-bound peer would travel 727'. That is 541' (a tenth of a mile) farther to reach the same location. A new elevator, close to the building's main entrance, would help eliminate this discrepancy.

RESTROOMS

No ADA-compliant restrooms exist in the Potter Center. Additionally, doors leading into restrooms do not have the necessary adjacent area to allow a wheelchair-bound student to operate the door. Restrooms on the second and third floor contain a stall with the width to accommodate ambulatory accessible use (38" wide), but the 4'-6" depth of these stalls is not compliant with the required 5' minimum.

Without significant restroom renovations, disabled students will likely need to leave the building to use the restroom.

Recommended Accessibility Upgrades

In order for the Potter Center to house the Disability Resource Center, we strongly recommend accessibility upgrades be performed at a minimum on and around areas that would serve the DRC, but ideally throughout the building. The following upgrades should occur where applicable:

- ADA-compliant door handles
- ADA-compliant signage
- ADA fire system notification
- Proper spacing around doors
- Proper clearance requirements in bathrooms
- Proper door closure forces
- Proper counter heights
- Proper outlet and switch heights

ARCHITECTURAL ASSESSMENT

Though in respectable condition for an over 40-year-old structure, the Potter Center has had very few improvements or cosmetic upgrades over its lifespan and aesthetically appears quite dated.

The building is generally designed in the "brutalist" or "brick brutalist" style popular for university, government, and high-rise housing projects from the 1950s to the mid-1970s. The façade is a combination of large areas of unadorned, blank, brick and unbroken, repetitive facades of narrow 5'-4" precast concrete panels. The panels themselves feature very deep protrusions and offer quite a bit of depth to the façade. However, the continuous repetition of the panels tends to temper the visual interest created by their depth and instead lends the precast areas a sense of monotony. The building's most visually intriguing exterior feature are its protruding brick stair towers with curved outer walls.

On the interior, the building's structural frame has the potential to be a highlighted feature. Its two-way concrete beam system (essentially a large waffle slab) is rather attractive. The deep beams running in perpendicular directions give the ceiling a highly textured aesthetic. Unfortunately, the slabs are currently hidden behind dropped acoustic ceilings in most locations. The floor system, in addition to the generous floor-to-floor heights, creates a space with the potential to become an attractive interior. The terrazzo flooring along corridors on the first floor is also an asset. Terrazzo is a very tough and durable flooring material that can be refinished and restored to nearly as-new condition.

With the exception of a row of seven large windows along the first story of the south face, window openings generally occur within the precast panels and are thus quite small at 5.4 sft each. The small window size, in conjunction with a 1'-4" precast concrete overhang directly above each window, allows only a limited amount of daylight into the building's interior. This, in combination with very limited corridor glazing, results in an interior that lacks natural light. Replacing areas of precast panels with glazing and opening up corridor walls would certainly allow more daylight into areas of the building; however, achieving a desirable amount of daylight across the entire floorplate will be complicated by the building's geometry. An "ideal" footprint for a well day-lit building is approximately 60' deep in the north-south direction and longer across the east-west direction. The Potter Center, however, is approximately a 160' square. Without cutting an internal atrium through the core of the building, it would be unreasonable to expect high levels of daylight to be present near the core, even after a substantial renovation.

The dated interior and exterior appearance of the building is not likely to affect the DRC or Central Testing Center; however, one could argue it may be a hindrance to the objectives of the College of Engineering. The COE departments slated to move into this building are student-centric departments focused on recruiting students to the college and encouraging the next generation of engineers. Engineering is a field of study focusing on innovation, research, and technology. This message may be diluted when delivered to potential COE students in a facility that projects an image of being outdated.

STRUCTURAL ASSESSMENT

Based upon our review of existing drawings, we were able to ascertain information defining the structural systems for the building.

During our walkthrough of the building, we visually verified to the greatest extent possible the existing structural systems. In general, the building structure appeared to be in good condition and generally as depicted in the original construction documents. In general, the structural system (both gravity resisting and lateral resisting) is intended to remain unchanged throughout the building as a result of the renovation of the facility.

FOUNDATION SYSTEM

The building has a full basement below its ground floor footprint. The basement is bounded by reinforced concrete walls supported on continuous reinforced concrete wall footings. The interior concrete columns are supported on a system of shallow reinforced concrete spread footings. The basement has a concrete slab on grade for its entire footprint.

FLOOR AND ROOF FRAMING SYSTEM

The floor and roof system is framed with a reinforced concrete waffle slab. The waffle slab is supported at regular intervals by reinforced concrete columns that continue down to the basement foundation level. At column locations, the concrete waffle slab construction transitions to solid slab.

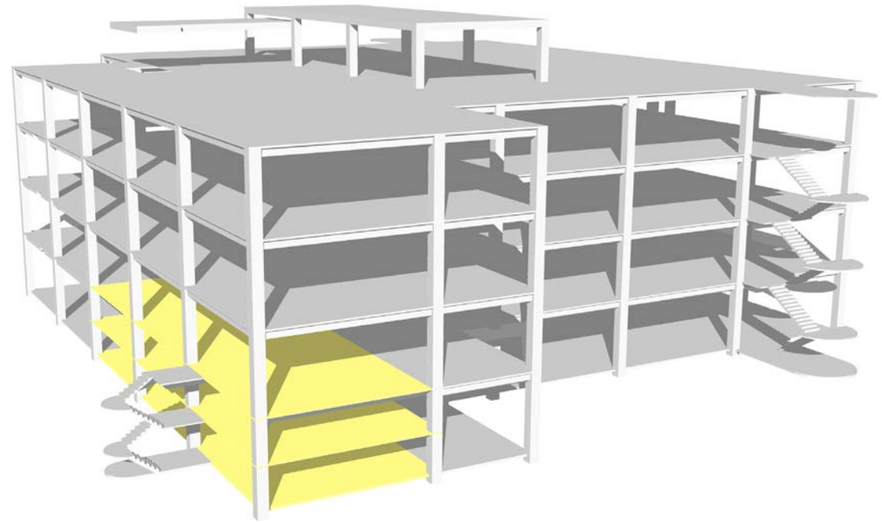
Special framing exists in the southwest corner of the building from the foundation level up to the second-floor level. In this area, steel framing exists that serves as library stacks. The steel structure frames the first-floor level as well as an intermediate level that occurs between the basement level and the first-floor level. The concrete waffle slab structure that occurs typically throughout the building does not occur at the first-floor level in this location.

LATERAL SYSTEM

Although it could not specifically be identified on the existing plans nor definitely confirmed during the walkthrough, it can reasonably be assumed that the lateral force resisting system for the building is accomplished through reinforced concrete shear walls at each of the stair shafts in the building (four total).

EXISTING MEZZANINE LEVEL

There is currently a steel stack framing system in the southwest corner of the building that partially vertically splits the 15’-4”-high basement level into two shorter 7’-8” floors. The headroom in these floor levels is approximately 7’-5”, with sprinkler heads at 7’-0”. These dimensions are lower than the 7’-6” head height requirement permitted by code. Therefore, the mezzanine level and basement area under the mezzanine level cannot be utilized if left unchanged. Removing the mezzanine framing would allow the basement level underneath the mezzanine to be utilized; however, because the framing system also infills an opening on the first floor, the first-floor area would be lost and there would not be a net gain of area. By removing the mezzanine and infilling the 3,000-sft opening on the first floor, the project would gain the square footage on the basement level.



EXISTING MEZZANINE LEVEL

CIVIL ASSESSMENT

The building's main entrance is currently located in the southeast corner of the building. This entrance's elevation is substantially above the grade of the adjacent alley to the east as well as the current hardscape on the south side of the building.

Access is gained to the main entrance from the surrounding lower grade by a series of stairs at the southeast corner of the building as well a ramp system that is located west of the stairs. There is currently no designated ADA-accessible parking areas that serve the building.

To improve the access to the main entrance of the building, ADA-accessible parking should be considered near the main entrance of the building. This parking would preferably be located adjacent to the ramp system that leads to the main entrance or as parallel parking with a clearly defined accessible route to the ramp system. The feasibility of providing the ADA-accessible parking will need to be studied in further detail with the current improvements being constructed for the Centennial Mall Streetscape and Utility Tunnel Repair 2016 project. Analysis of the existing grades and available space along the existing alley to provide parallel parking should also be investigated to ensure the best accessible option is pursued.

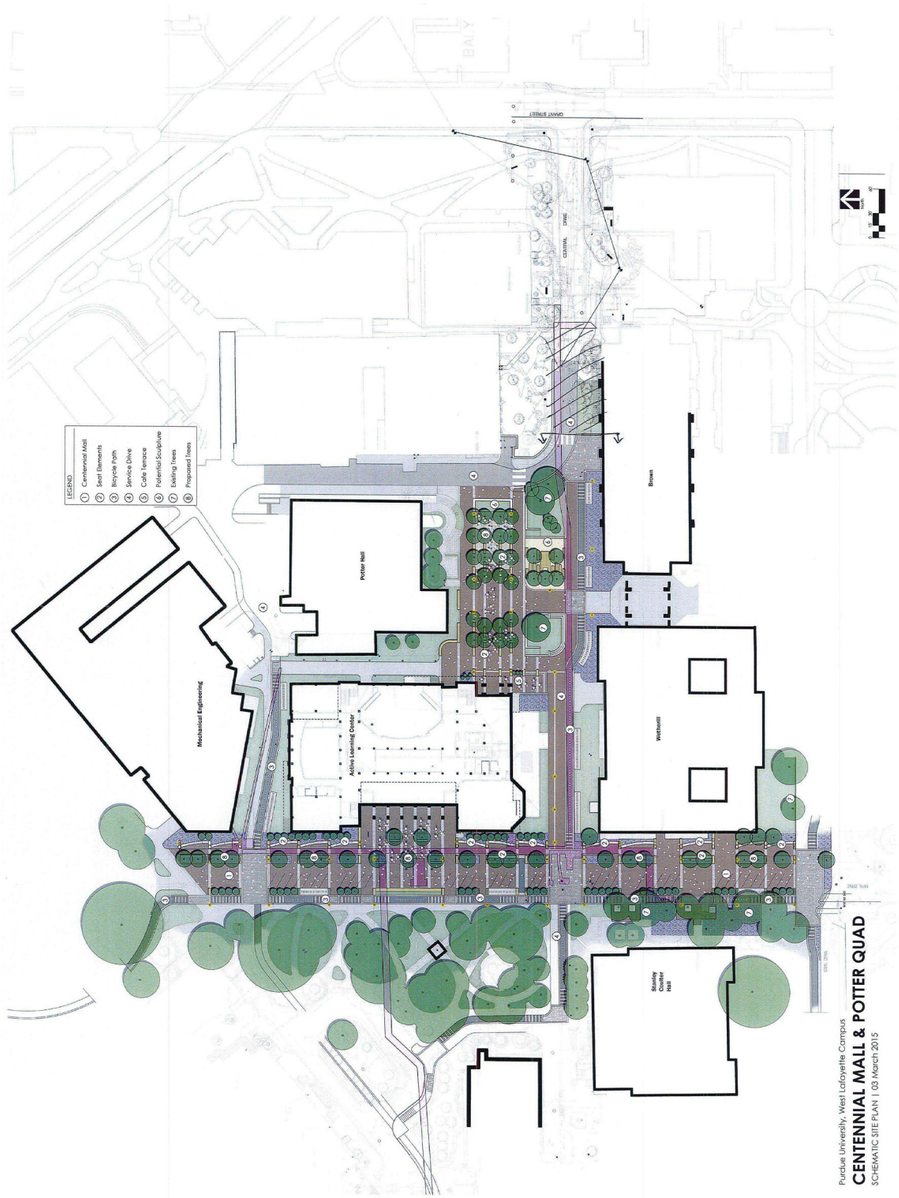
CENTENNIAL MALL SITESCAPE AND UTILITY TUNNEL REPAIR

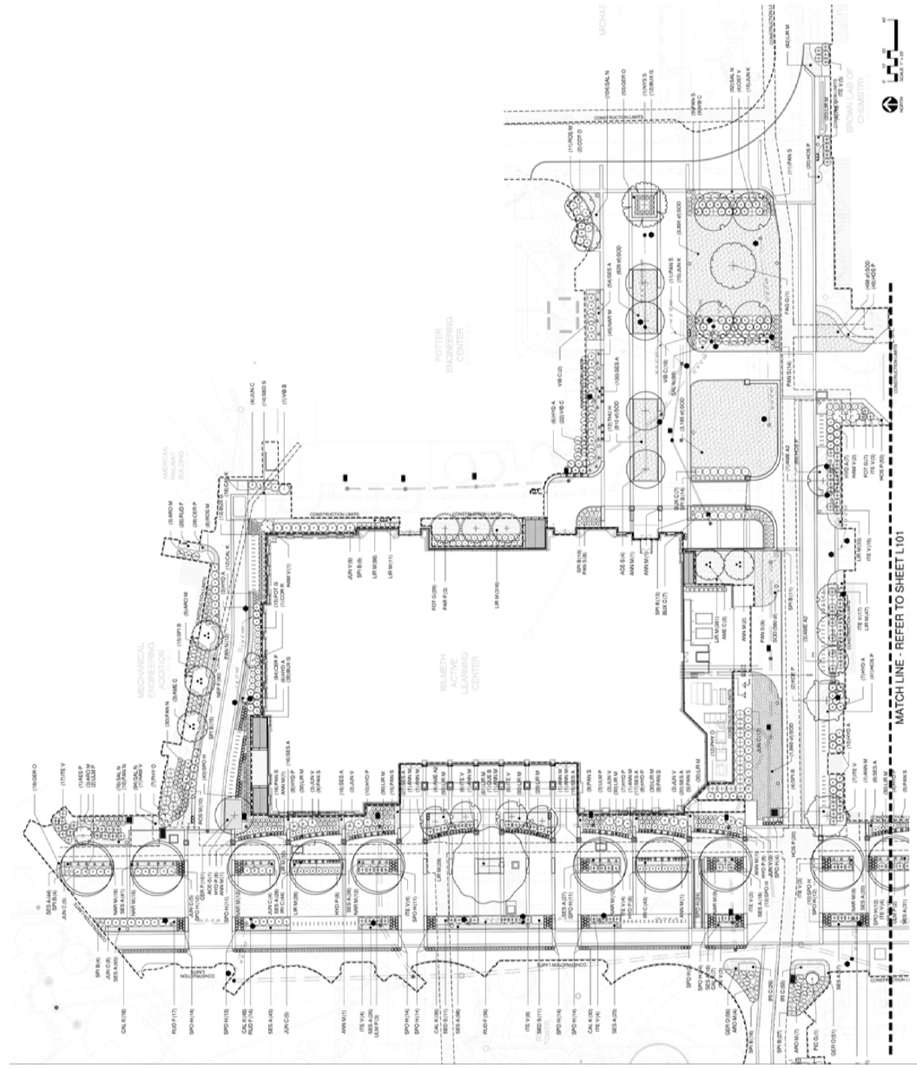
This site and landscaping project will occur immediately south of the Potter Center and around the Wilmeth Active Learning Center, immediately west of Potter. It will fully reimagine Centennial Mall.

As a whole, the proposed design treats the Potter Center as a background building and is designed solely in deference to the Active Learning Center. Though this project directly affects the Potter Center's "front yard," it generally ignores the Potter building. There is some attention paid to installing new landscaping at the southeast stair, which is currently the building's main entrance. This is especially concerning with regard to the Disability Resource Center, a department whose sole purpose is to accommodate students with special needs, since the main entry stair is inaccessible. Students who need to use a ramp to access Potter from the south must travel to the west, to a ramp hidden in a small break in the landscaping. Students who cannot use the stairs are forced to travel a minimum of 138' and a maximum of 274' farther than their peers.

It is the strong recommendation of this study that if the Potter Center is ultimately remodeled to create a new home for the Disability Resource Center, then accessibility to the building should be simultaneously addressed so all students can use a common entry path to the building. If the primary intent of the relocation of the DRC is to move it to a more accessible location, that goal will not be met without a concurrent exterior renovation.

The lack of integration of Potter's circulation flow into the Centennial Mall project should be concerning for the College of Engineering as well. The primary goal of the COE's reimaging of Potter is to house all of the student-centered engineering departments under one roof. Many of these departments are focused on recruiting and encouraging the next generation of engineers—functions that will actively benefit from student foot traffic and a high degree of visibility to campus. These goals are not supported by the current Centennial Mall design.


LEGEND
1 Centennial Mall
2 Seat Elements
3 Bicycle Path
4 Service Drive
5 Cafe Terrace
6 Potential Sculpture
7 Existing Trees
8 Proposed Trees
Potter Hall
Mechanical Engineering
Active Learning Center
Wetherill
Stanley Coulter Hall
Brown
GRANT STREET
CENTRAL DRIVE
Purdue University, West Lafayette Campus
CENTENNIAL MALL & POTTER QUAD
SCHEMATIC SITE PLAN | 10 March 2015


MATCH LINE - REFER TO SHEET L101

MEP ASSESSMENT

MECHANICAL SYSTEMS, HVAC

- The building uses campus steam and chilled water for heating and cooling.
- Beginning in circa 2000, a heating water system was added to support the VAV terminal unit replacement projects.
- Between approximately 2000 and 2013, all of the original terminal units were replaced with VAV reheat terminal units.
- A list of obsolete VAV terminal unit controllers was obtained, and in general, all but the second-floor units are obsolete.
- The building is served by eight air handlers.
- In general, the casings of all of the air handlers observed were in reasonably good shape. No observations could be made on the condition of the fans and the coils.
- There are three main air handlers.
  - ACB-1 serves most of the basement and first floors.
  - ACP-7 serves the third floor.
  - ACP-8 serves the northern portions of the first, second, and third floors.
  - During a site visit, it was observed all three of these air handlers were running in the 45 to 55 Hz, meaning there is little additional capacity available. This was between 8 and 9 AM on a day when the outside air temperature was approximately 25° F and the loads on the systems should not be near peak conditions.
- There are four small air handler/fan coil units that serve small functional areas.
  - ACB-2 provides perimeter heating and cooling for the first, second and third floors.
  - ACB-3 serves primarily the northern portion of the basement.
  - ACB-21 is a 2 to 3-year-old unit that serves labs in the basement adjacent to the mechanical room.
  - AC3-17 serves a suite of small rooms.
- In 2013, a study of the controls systems was conducted, and most of the energy savings options recommended were accomplished leaving little room for additional system temperature control improvements.
- A spot check of supply duct air velocities and static pressure drop per 100' indicates the ductwork was most likely sized using the static regain method. This generally leads to smaller ducts with higher operating duct static pressures and higher horsepower requirements, which are not acceptable under the current energy code.
- The existing air handler systems are intertwined, making it difficult to repurpose areas of the building without affecting more than one system.

MECHANICAL SYSTEM, PLUMBING

- The sanitary waste system incorporates both a gravity drain main and a lift station.
- The building is served by a 3" domestic water line at the northwest corner of the building.
- The domestic hot water system is served by a water softener in the basement mechanical room and utilizes a vertical storage water tank with a steam-fired tube bundle. The system has a return water pump.

ELECTRICAL SYSTEMS

- The transformer vault is located in Room B049, with transformers L-1 and P-1 (primary 12.47 kV, secondary 208V) and P-2 (primary 12.47 kV, secondary 480V).
- The switchboards, ATS, and emergency panel are in Room B048. This consists of 208V distribution switchboards M-L-1 and M-P-1, the 480V distribution panel MSB-1, and panel EM-1.
- Fed from M-L-1 are four bus ducts, one each to the northwest, northeast, southwest, and southeast corners of the building. From the basement, these ducts run vertically and serve the 208/120V distribution and/or lighting and receptacle panels on each floor.
- Room B046 has a motor control center MCC-G-1 that serves mechanical loads and a 208/120V panel L-G-23 for serving the lighting, receptacle, and other smaller loads in this area.
- B003, B023, and B043 each have one 208/120V distribution panel with fused switches and one 208/120V lighting and receptacle fused-style panel board. Basement Room B075 has one 208/120V lighting and receptacle fused-style panel board.
  - SPD-G-SE and L-G-SE in B003
  - SPD-G-NE and L-G-NE in B023
  - SPD-G-NW and L-G-NW in B043
  - L-G-SW in B075
- Room 143 has one 208/120V distribution panel with fused switches and one 208/120V lighting and receptacle fused-style panelboard. Rooms 103, 123, and 175 each have one 208/120V lighting and receptacle fused-style panel board.
  - L-1-SE in 103
  - L-1-NE in 123
  - SPD-1-NW and L-1-NW in 143
  - L-1-SW in 175
- Rooms 203, 223, 243, and 275 each have one 208/120V distribution panel with fused switches and one 208/120V lighting and receptacle fused-style panel board.
  - SPD-2-SE and L-2-SE in 203
  - SPD-2-NE and L-2-NE in 223
  - SPD-2-NW and L-2-NW in 243
  - SPD-2-SW and L-2-SW in 275
- Rooms 303, 323, 343, and 375 each have one 208/120V distribution panel with fused switches and one 208/120V lighting and receptacle fused-style panel board.
  - SPD-3-SE and L-3-SE in 303
  - SPD-3-NE and L-3-NE in 323
  - SPD-3-NW and L-3-NW in 343
  - SPD-3-SW and L-3-SW in 375
- Room 401 has a motor control center MCC-PH that serves mechanical loads and 208/120V panel L-PH-1 for serving the lighting, receptacle, and other smaller loads in this area.

TELECOM SYSTEM

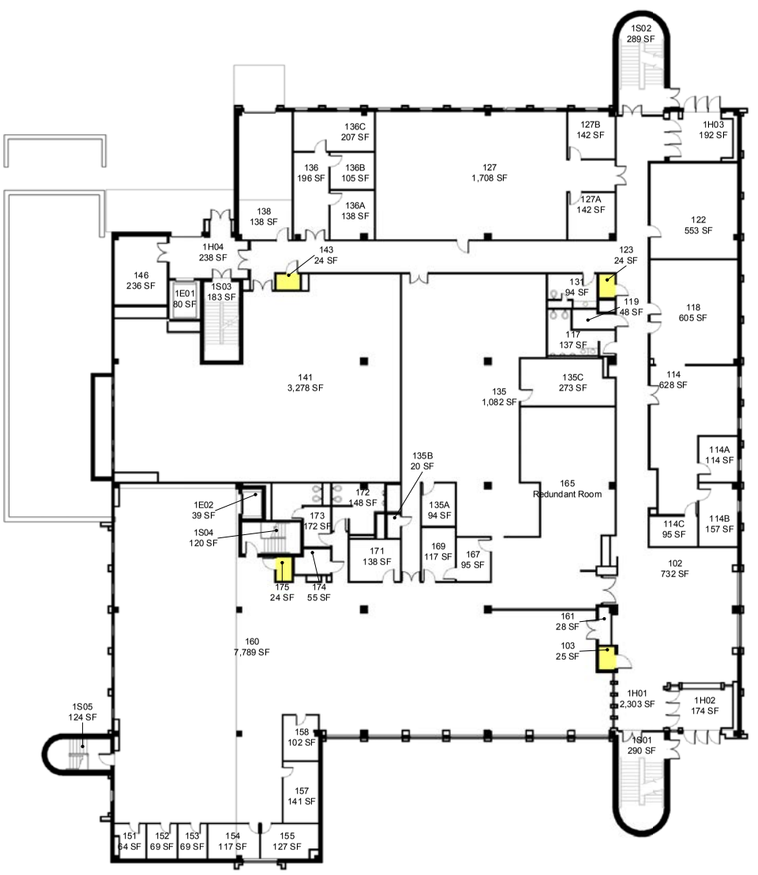
- BDF is located in Room B045. This room was very warm when we did our site survey. We have since confirmed with ITaP that the BDF is cooled by a fan coil unit in need of maintenance.
- The building is currently fed with a 250-pair copper cable from the Tel building (CENT02, 1301-1550 with bad pairs on 1548, 1526, 1725, and 1726) as well as a 24 single-mode fiber fed from the Tel Building (CENT FO1204, 97-120)
- IDF is located in Room 401.
- There is a telecom switch with some wall-mounted patch panels in Room 175.
- All other basement, first, second, and third floor rooms in the northeast and southwest corners of the floors that are noted as IDF on the existing plans are just pass-through spaces with no telecom equipment. These spaces are actually electric closets.
- Existing cabling is CAT 5e.
- Potter has a 600pr copper cable and two 12-strand SM fiber-optic OSP cables serving the building.

FIRE ALARM SYSTEM

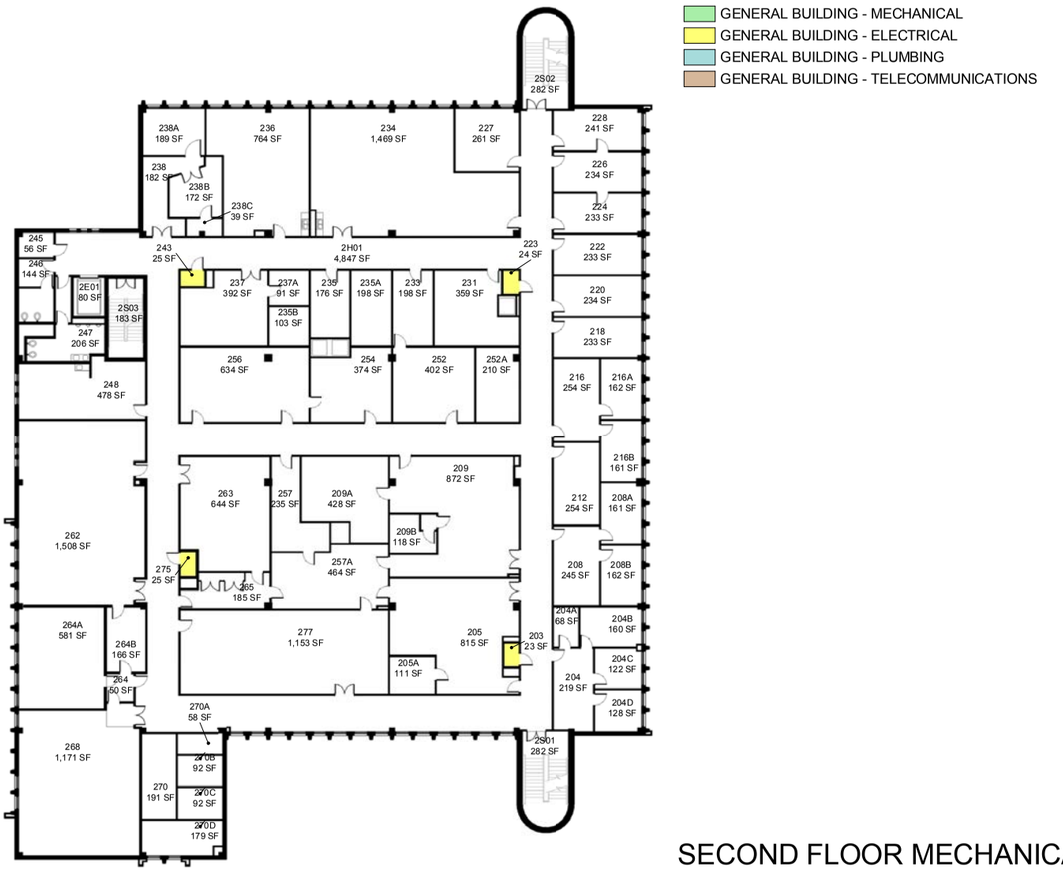
- Appears to be a fairly new Simplex 4100U-ES system installed.



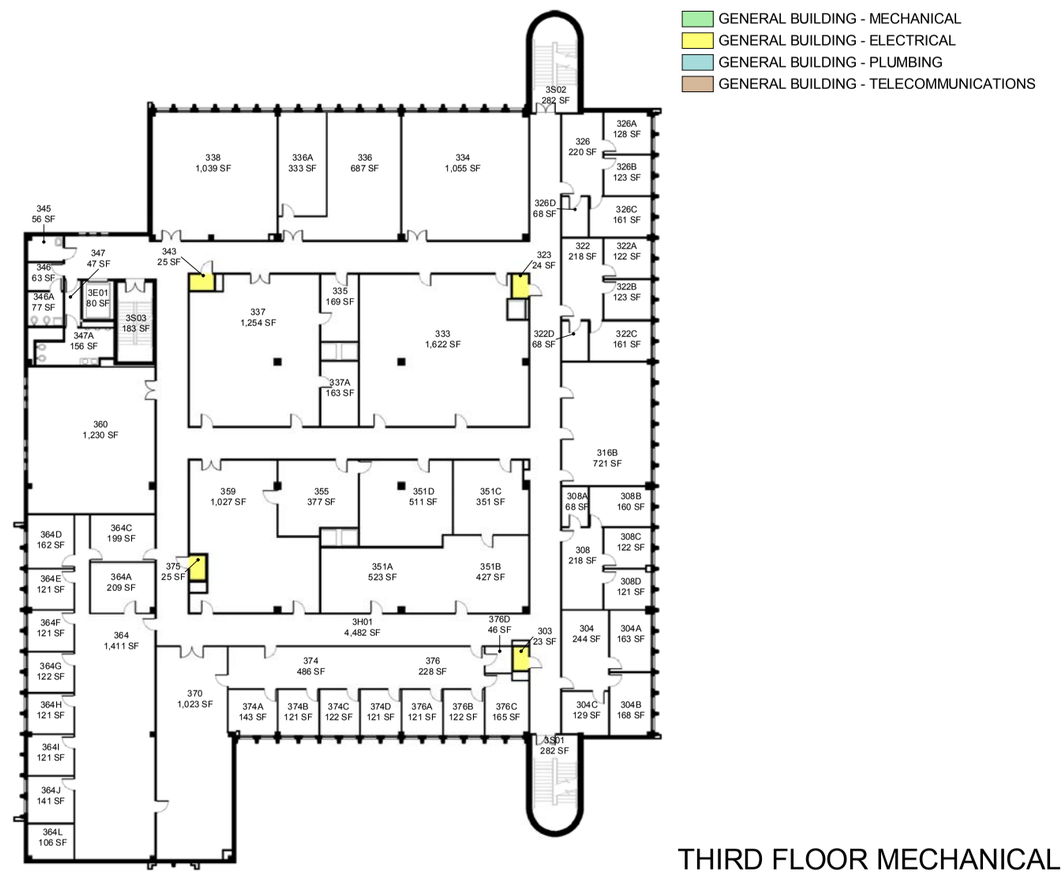
LOWER LEVEL MECHANICAL



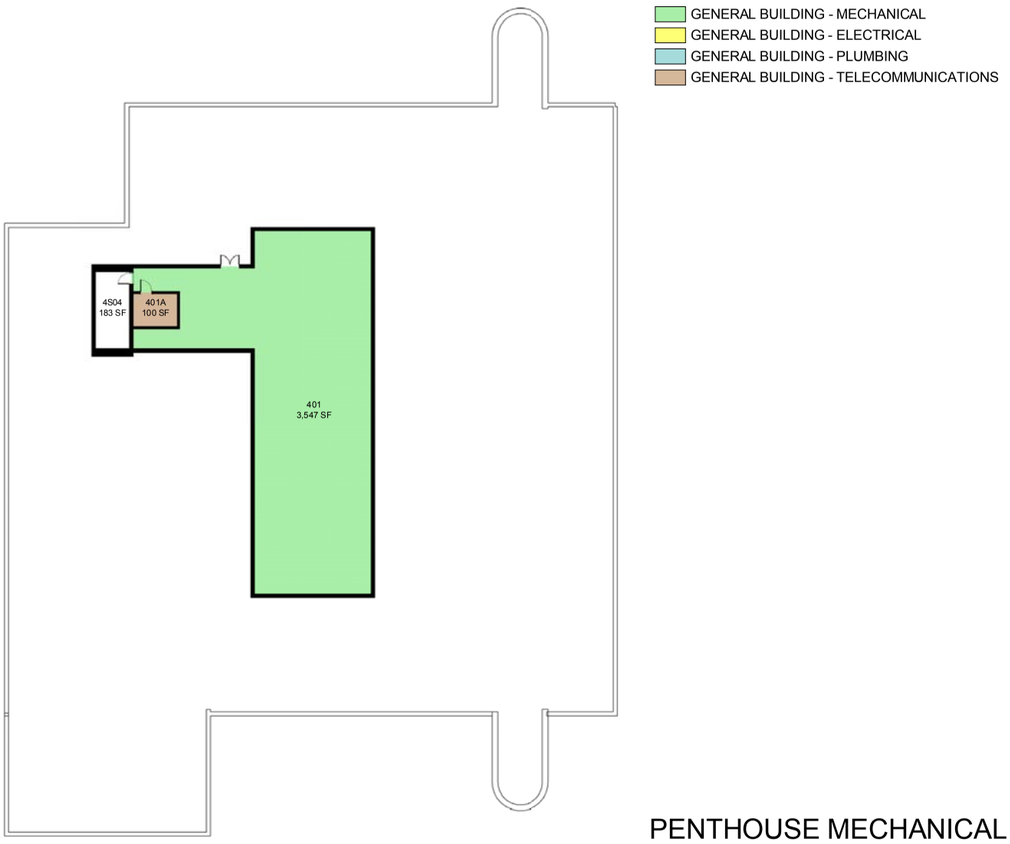
FIRST FLOOR MECHANICAL



SECOND FLOOR MECHANICAL



THIRD FLOOR MECHANICAL



PENTHOUSE MECHANICAL

CENTRAL TESTING CENTER

The goals of the CTC are to create a new campus-wide resource that will present faculty with a new option for testing and increase testing flexibility for courses across campus; to increase cost control by efficiently combining testing services and departments; to increase efficiency of resources through greater utilization of staff, testing equipment, and testing space; to increase academic integrity by creating purpose-built space to effectively proctor exams administered by a robust and specifically trained staff; and to greatly increase ease of use both for students and faculty by creating a single destination for testing needs.

CURRENT TESTING CENTER USAGE

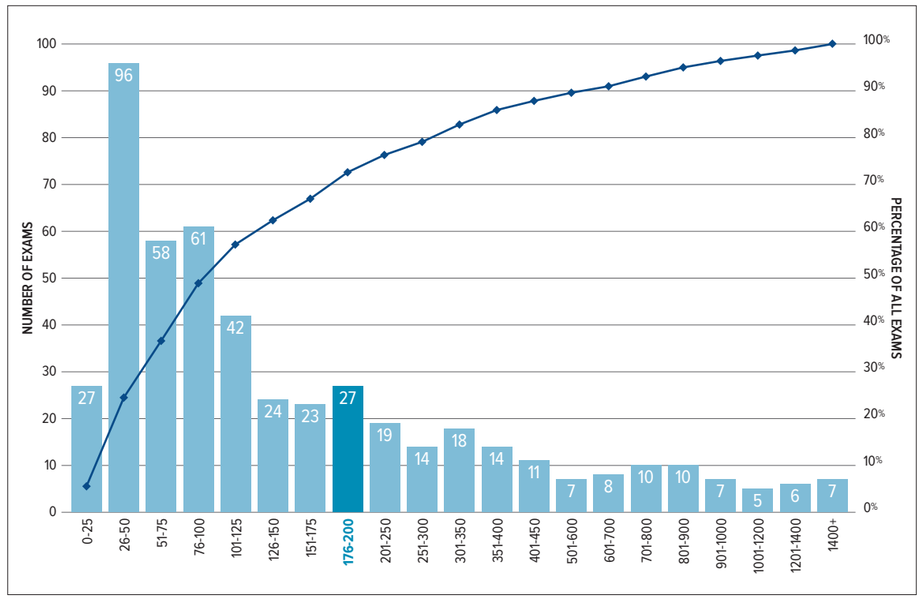
The four testing centers being consolidated into the Potter Center facility have a total of 101 testing seats. However, the ultimate goal of the testing center is not solely to create a larger and more efficient aggregated testing center, but to create a new resource for testing that can be used by instructors across campus. Therefore, the testing center has been sized to serve campus as a whole.

Testing Center	Seats	Tests/Day*
Disability Resource Center	24	15
Purdue Testing Center	16	30
Digital Education	25	20
Foreign Language Placement	36	30
SUM	**101**	**95**

* Foreign Language Placement testing does not occur year round.

Beyond the need to bring units together is the need, as shared by University leadership, to create better conditions for exam delivery at Purdue. A task force on this topic, under the leadership of the director of Digital Education, has been meeting for about a year. The result has been to encourage the bringing together of testing units and to find ways to address University concerns about test integrity and proctor services. One specific result is the conclusion that a contained, flexible, and carefully managed testing space that can accommodate up to 200 students at one time would be a definite asset to faculty and instructional staff. This new center will additionally support the expansion of proctored testing for Purdue's flipped/hybrid classes that are redeveloped as part of IMPACT and the growth of online courses likely to flow from the new strategic plan for Digital Education. The center will also reduce the demand for academic and non-academic spaces during periods of volume testing. Furthermore, faculty are presented with a facility where their exams could be delivered in multiple ways, using the testing center or their own personnel or both. The center will likewise serve as a crucial asset during the 25 days in the summer during STAR and BGR.

The following chart illustrates the number of evening exams given on campus in the fall of 2014. Evenings are the time of day in which most tests are administered. As evidenced by the chart, a testing center with between 176 and 200 seats would serve approximately 74 percent of campus.



CURRENT TESTING CENTER FACILITIES AND CONDITIONS

There are currently four small testing centers spread throughout campus—Foreign Language Placement, Digital Education, the Disability Resource Center Accommodated Testing Function, and the Purdue Testing Center. Each center focuses on a specific testing need and has a dedicated staff. Each also has a unique way of administering their specific test(s), as well as specific rules and regulations for the administration of that test. The intent of the Central Testing Center is to combine these functions into one purpose-built location. Analysis of the four testing centers follows here.

* Items marked with this symbol appear to be the most stringent of this line of requirements and will be labeled as a "controlling" requirement. The testing center, in part or in whole, will then be designed toward meeting this requirement.
** This area presents an interesting synergy with some of the functions housed within the College of Engineering that focuses on introducing students to the discipline of engineering and may benefit from the exposure to incoming freshman.

Foreign Language Placement Testing

The goal of the foreign language testing center is to effectively place students in foreign language courses appropriate for their current comprehension of that foreign language.

Physical Test Administration Requirements

- Windowless and distraction-free environment
- Computer at each testing station
- Large line for queueing students during peak demand that is acoustically separated from the testing center*
- Gathering area for students waiting for others to complete testing**
- Pre-screening area to conduct individual interviews with students prior to testing*
- Semi-private post-testing area where students are debriefed on their test results*
- Storage area for headphones

Testing Personnel

- Three TAs are used during the year
- Five TAs are used during STAR
- Some TAs must be specially qualified to effectively speak the language being tested in order to verify that students are sitting for the correct test*

Testing Load

- Number of Testing Seats: 36
- Total Number of Students Tested Per Regular Session: 90 to 100
- Total Number of Students Tested During STAR: 150 to 300 per day for 19 days
- Typical Duration: 15 to 60 minutes

Current Testing Hours

The testing lab is open during five sessions per year.

- The month of August
- Three days during November
- Two days during January
- Three days during April
- STAR - Last two weeks of June through first two weeks of July (19 days) from 10 AM to 3 PM

Current Conditions

The Foreign Language Testing Lab currently operates out of a standard computer lab in Stanley Coulter Hall, Room 231, which is repurposed for testing during five sessions per year. Computers are not purpose built for testing. TAs must log in and manually set up each station before testing begins. Tests are web based. There are no storage lockers for jackets and backpacks; therefore, students bring all their items with them to the testing center. The testing stations do not have partitions between students. Both the pre-test screening interview and post-test debriefing are conducted inside the testing room.

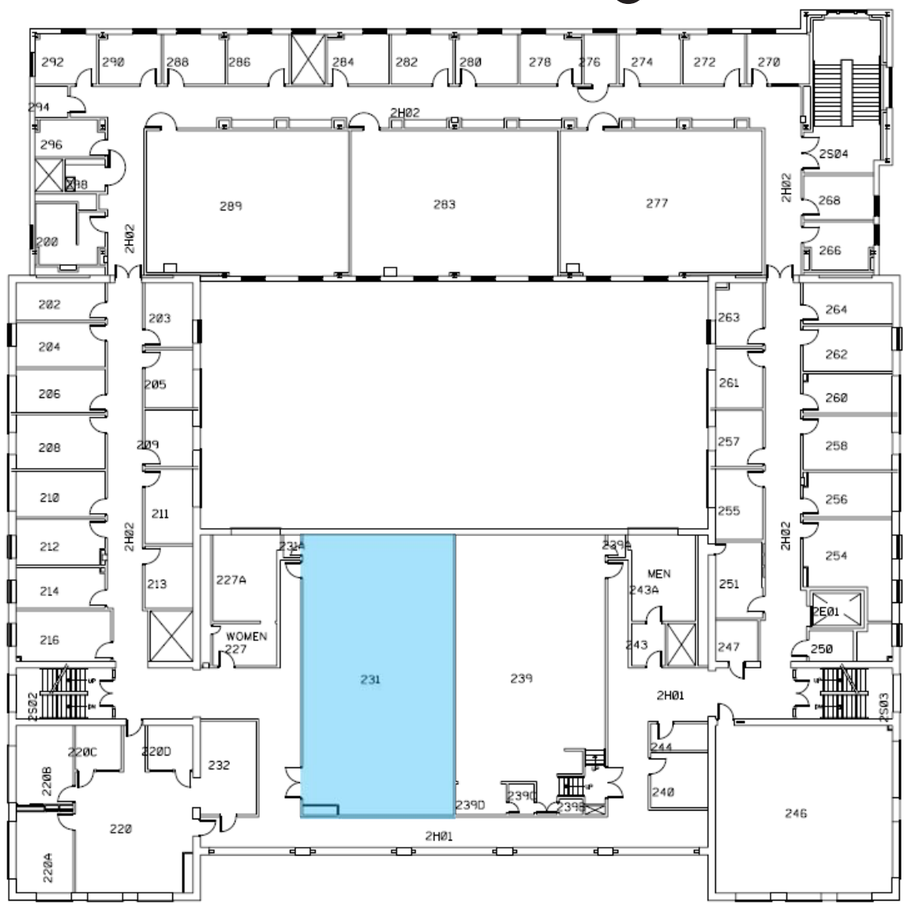
Current Sequence of Operations for Testing

- Students do not schedule exams in advance; they simply arrive at the testing site carrying a permission form (Form 231).
- Students line up in the hallway leading up to the testing center. The line can be very long, and the noise created by large numbers of students talking can be distracting for students taking the test, as well as for other occupants of the building.
- Students sit down at a table to go through the screening process. Note that the testing staff feels strongly that this interaction with students is critical to the efficiency and academic rigor of the testing.
- Each student hands TA Form 231 and their student ID for TA to verify their identity.
- The duration of the screening process is 7 to 10 minutes.
- Once the student has finished the screening process, they are directed to sit at an open computer to take their exam. It is up to the student to choose the correct exam. Students bring jackets, backpacks, and any other materials they have with them to the test station. TAs monitor students in person to discourage cheating.
- The duration of an exam is 15 to 60 minutes. Tests do not have scheduled breaks. Due to the short duration of the test, it is very rare for a student to ask to leave to use the restroom. Some tests require headphones for listening comprehension.
- Students receive results immediately following completion of the exam.
- Once a student completes the exam, the student raises a hand and a TA walks over to explain the results.
- Students exit the exam and often wait for others to complete their exam.

Potential Areas for Improvement

- Acoustically separated waiting area for students
- Mechanism to scan student IDs prior to testing
- Ability to scan Form 231 on site during the screening process
- Ability to test more frequently throughout the year

Current Floor Plan STANLEY COULTER HALL (SC) - SECOND FLOOR



Digital Education Testing Center

The Digital Education Testing Center is an on-campus proctored testing site that administers a variety of paper-based and computerized exams. All testing is scheduled ahead of time by individual students. Tests administered in this center are for Purdue-based distance learning courses, as well as CASA (Core Academic Skills Assessment) and Pearson Vue exams.

Physical Test Administration Requirements

- Some requirements vary by test. When proctoring tests for professors as make up exams, the requirements are dictated by that professor.
- Lockers for student item storage
- White noise system in testing area
- Business office in close proximity to handle cash or check payments by students for certain tests*

Testing Personnel

- One full-time staff

Testing Load

- Number of Testing Seats: 25
- Number of Students Typically Tested Simultaneously During the Week: 5-6
- Number of Students Typically Tested Simultaneously on Saturday: 20
- Typical Duration: 90 Minutes

Current Testing Hours

- Monday, Tuesday, Friday: 8:30 AM to noon and 1:30 to 4:30 PM
- Wednesday, Thursday: 10 AM to noon and 1:30 to 4:30 PM

Current Conditions

The Digital Education Testing Center is located in the lower level of the Stewart Center in Room G59. The test proctor sits outside the room and can view students through windows in the wall located approximately 4' above ground. The proctor can also monitor students through video cameras mounted in the ceiling. All materials students bring to the testing center are stored in lockers, unless the testing center has received specific instructions to allow students to bring items with them to the test. The testing room is currently not acoustically well isolated from the outside, and noise can cause an issue. A white noise system is installed in the ceiling to create background noise. All testing is taped. The video is stored for 30 days.

Current Sequence of Operations for Testing

- Student arrives within 15 minutes of their scheduled testing time.
- Student shows ID to the staff for identity verification.
- Student signs login form.
- Staff assigns student a locker to store any materials not permitted in the testing room. Student keeps the key during the exam.
- Students must pay a fee for certain tests. Most pay fees online. Students with cash or checks are sent to the business office and then return to the testing center with a receipt.
- For certain tests, students must have their picture and thumbprint taken.
- The student is directed toward a specific station to take their exam. Only certain computers can administer CASA tests, and separate computers can administer Pearson tests. The computer setup is dictated by the testing company. Paper exams account for 40% of tests; 60% are online.
- Exams do not have scheduled breaks. If students ask to leave to use the restroom, the clock continues running on their exam. It is very rare for students to ask to leave during an exam.
- Once the student completes an exam, he/she exits the room.
- Each student signs the logout form.
- Student retrieves materials from an assigned locker and leaves the testing center.
- Completed paper exams are then mailed, scanned, or picked up by the professor.

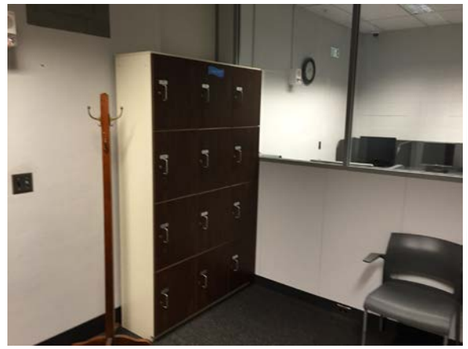
Potential Areas for Improvement

- Greater acoustic isolation for the testing room
- Better view into the testing room for the proctor

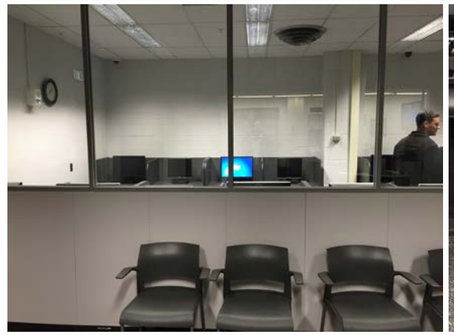
Online Compliance

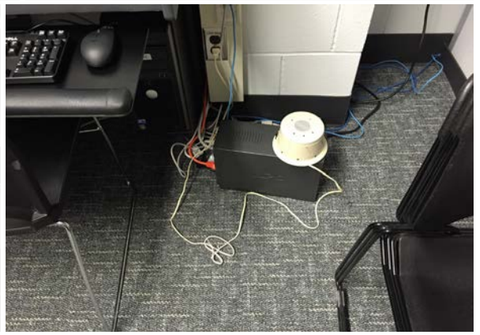
This new unit is being created per a requirement set by Purdue's vice president for Ethics and Compliance, who will oversee compliance with digital materials. The proposal for the unit is currently under review by Purdue's in-house legal council. The unit will be housed within Digital Education and be led by an associate director of Digital Education. While the associate director position is funded and was recently filled, the remaining positions are yet to be funded. These staff (and the corresponding demand for space) are therefore new and not connected with the existing units. As part of this project, the staff for this unit will be accommodated. This unit will work in conjunction with the Disability Resource Center, and locating these offices in proximity will enhance collaboration and increase efficiency.

Current Photos

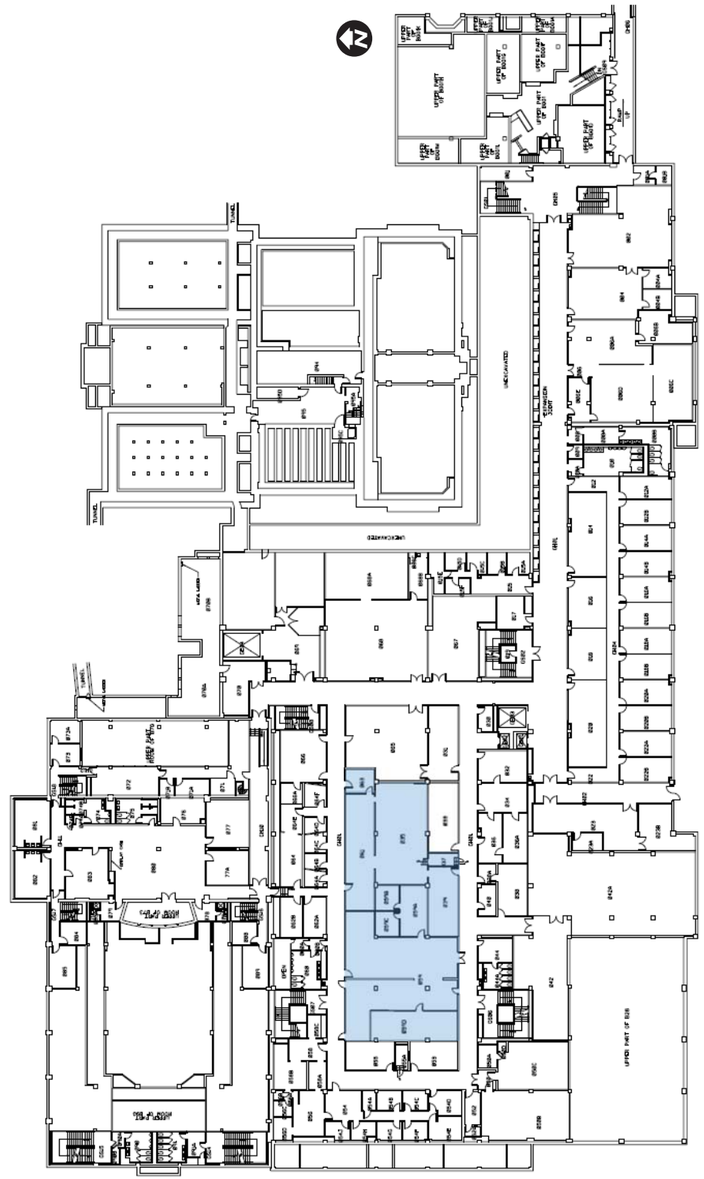








Current Floor Plan STEWART CENTER (STEW) - GROUND FLOOR



Disability Resource Center (Accommodated Testing Function)

The goal of the accommodated testing function of the Disability Resource Center is to create an environment that accommodates students with special needs when taking exams during typical coursework.

Physical Test Administration Requirements

- Windowless, very quiet, distraction-free environment
- Lockers for student item storage
- Testing stations must be clear of all items, including computer monitors*
- Private, isolated testing rooms (used for certain students who require extreme isolation from distraction and for students who must take a test with another person working as a scribe/reader)*

Testing Personnel

- Two full-time staff

Testing Load

- Current Number of Testing Seats: 24
- Number of Students Typically Tested During the Semester: 2,500
- Number of Students Typically Tested During Finals Week: 650

Current Testing Hours

- Weekdays from 8 AM to 8 PM

Current Conditions

The Disability Resource Center and its associated testing center is located in Young Hall, Room 830. The testing function of the DRC is among the critical functions provided by the center. One-third of all students registered with the DRC use the testing center. Unlike the majority of tests given at other test centers, testing at the DRC is nearly all paper-and-pencil testing. The primary accommodation provided by the DRC is to allow more time for test taking. Tests are scheduled by students. Students primarily take tests in a large room with testing stations divided by study carols, similar to the computer testing centers. A small number of students requiring additional isolation or assistance from a scribe or reader use private testing rooms. Up to 12 private testing rooms can be used at one time. The testing center's greatest demand occurs during finals week. During this time, the DRC sets up four testing locations across campus, and proctors administrate approximately 650 exams.

Unfortunately, during this period, the demand for accommodated testing far exceeds the capacity of the DRC's testing space. The DRC makes use of libraries and other spaces to house accommodated exams. More importantly, the DRC staff must be augmented by volunteers and paid proctors from its parent unit (Student Success) and other units at Purdue. The Central Testing Center will create the ability for accommodated testing to occur in a much more efficient and controlled way and with a deeper overall testing staff (to oversee the inevitable volunteer proctors).

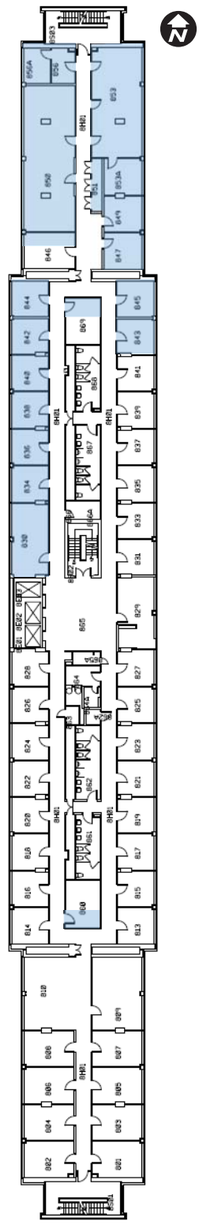
Current Sequence of Operations for Testing

- Student arrives at the DRC at the scheduled testing time.
- Student checks in with staff.
- Student shows ID to staff for identity verification.
- Staff assigns student a locker to store any materials not permitted in the testing room.
- Staff assigns the student a seat number where he/she goes to take their exam.
- Student leaves DRC.
- Completed tests are picked up from the DRC. A large amount of these tests then travel to IDP for grading.

Potential Areas for Improvement

- Need 10 private testing rooms
- Need 30 to 35 testing seats
- Video recording for private testing rooms

Current Floor Plan ERNEST C. YOUNG HALLL (YONG) - EIGHTH FLOOR



Purdue Testing Center

The goal of the Purdue Testing Center is to serve campus and patrons outside of campus by providing a variety of testing services. It is the only center on campus fully certified by national testing organizations and offer national registration and licensing exams.

Physical Test Administration Requirements

- Lockers for student item storage
- White noise system in testing area
- Secure server stored in a locked location to store testing video*
- Locked room for national test storage*
  - No windows permitted
  - Must have a hard lid ceiling
- Security Equipment*
  - Three screens for video monitoring
  - Three DVRs
  - Thumbprint scanner
  - Student card swipe
  - Area to take pictures
  - Signature pad
  - Palm vein scanner
  - Wand-type metal detector

Testing Personnel

- One full-time staff

Testing Load

- Highest Number of Students Per Day: 28 to 32
- Number of Testing Seats: 16

Current Testing Hours

- Monday, Tuesday, Friday: 8:00 AM – 5:00 PM
- Wednesday, Thursday: 8:00 AM – 9:00 PM

Current Conditions

The Purdue Testing Center is located in Schleman Hall, Room B42. Of all the testing centers on campus, the Purdue Testing Center has the most stringent testing requirements and accountability standards. This is driven by the standards required by the national testing companies. Videotaping of testing is critical, and footage is stored on a locked server. The tests given in this center can be quite long (some are six hours) and include mandated breaks. Due to the high number of students taking advanced credit exams, those are given outside the center, typically in Hampton hall, which seats 40 to 80 at a time.

Current Sequence of Operations for Testing

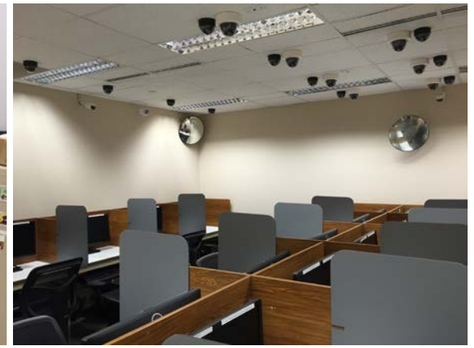
- Students line up to be checked in for testing.
- Three to four students at a time load their belongings into a locker.
- Some students are directed to read certain material before they are admitted.
- Students line up for admittance.
- Students go through security procedures required by that test (signature pad, palm vein scanning, thumbprint, picture taken).
- Proctor walks student to outside the testing room.
- Proctor pats the student down or wands the student.
- Student waits until a station is open and sits down.
- When student completes exam, he/she signs out and leaves the center.

Potential Areas for Improvement

- Break area with restrooms and vending machines inside the “security zone.”
- Secure restrooms should have multiple fixtures for quick use. Some students only receive an 8-minute break during their test.
- Need a larger testing center with 30 to 40 seats, possibly two 20-person rooms.

Current Photos



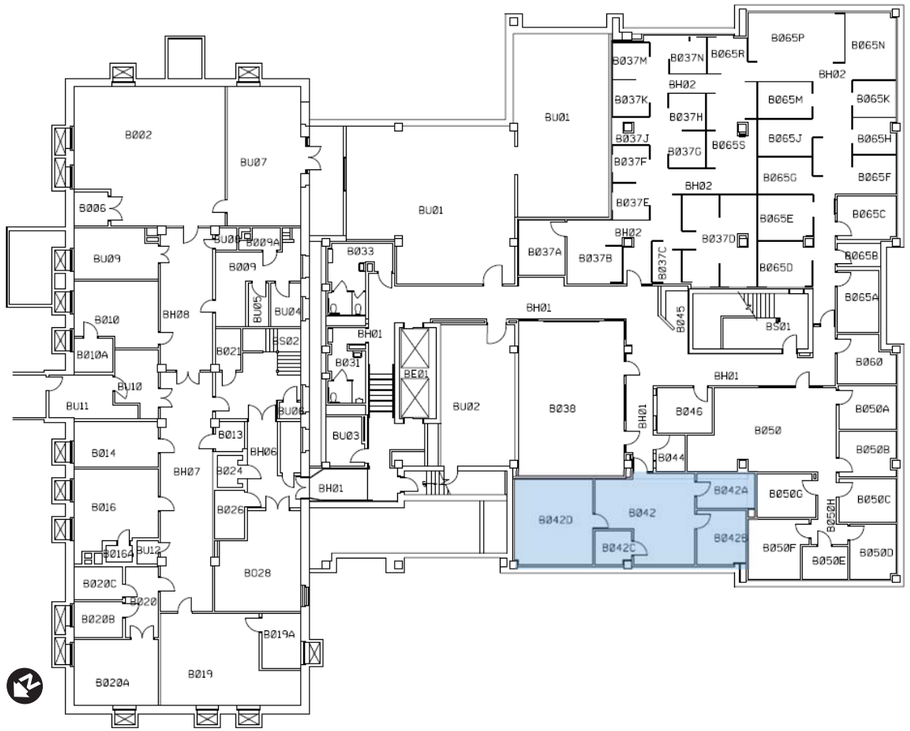








Current Floor Plan HELEN B. SCHLEMAN HALL OF STUDENT SERVICES (SCHL) - BASEMENT



SPACE PROGRAM

Position	Number of Personnel	Area Per Person	Total Area
HS Testing - Office	1	120	120
HS Testing - Video Proctor	1	90	90
HS Testing - Work Stations	4	90	360
Online Compliance - Private Office	1	120	120
Online Compliance - Shared office	3	120	360
Testing - Administration Counter	1	90	90
Testing - Director	1	120	120
Testing - Office	2	120	240
Testing - Video Proctor	1	90	90
Grand Total	**15**	**960**	**1,590**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
General Testing Stations	22.5	230	5,175
High Security Testing Stations	26	25	650
Lockers, 250	300	1	300
Lockers, 35	30	1	30
Private Testing Room	48	21	1,008
Queueing for 60	540	1	540
Secure Break Area	150	1	150
Secure Restrooms	64	6	384
Secure Storage	200	1	200
Storage	50	1	50
Testing Check-in	400	2	800
Testing Concierge	200	1	200
Grand Total	**2,030.5**	**291**	**9,487**
		TOTAL	**11,077**

CENTRAL TESTING CENTER DESIGN

The overall goal of the Central Testing Center design is to combine these existing testing centers in one purpose-built location. The testing center will also accommodate general “course” tests. In this instance, professors will direct entire classes to the testing center for administration of tests as required by that course. In general, the testing center will focus solely on testing as administered by computer. Professors administering traditional pencil-and-paper tests will be encouraged to continue to administer those tests in their classroom spaces. The exceptions to this rule will be for students with special needs and for make-up testing. In these instances, the center will accommodate pencil-and-paper testing.

The majority of tests administered in the center will be in service of a Purdue University course. However, the center will also handle specialized national licensing and certification exams, such as those currently offered through the Purdue Testing Center and the Digital Education Testing Center. These national exams are offered through outside testing vendors that give franchise locations the

ability to administer these tests providing they follow strict administration guidelines. Due to the stringent nature of these requirements, creating a secondary testing space for the administration of these “high-security” tests appears to be the most prudent course of action. Not applying these rules to the entire testing center will allow the larger “general” testing center a higher degree of freedom and flexibility. Therefore, as evident in the design of the space, the centralized testing center will have two distinct spaces—one for general testing and another smaller space for high-security testing.

CENTRAL TESTING CENTER SEQUENCE OF OPERATIONS

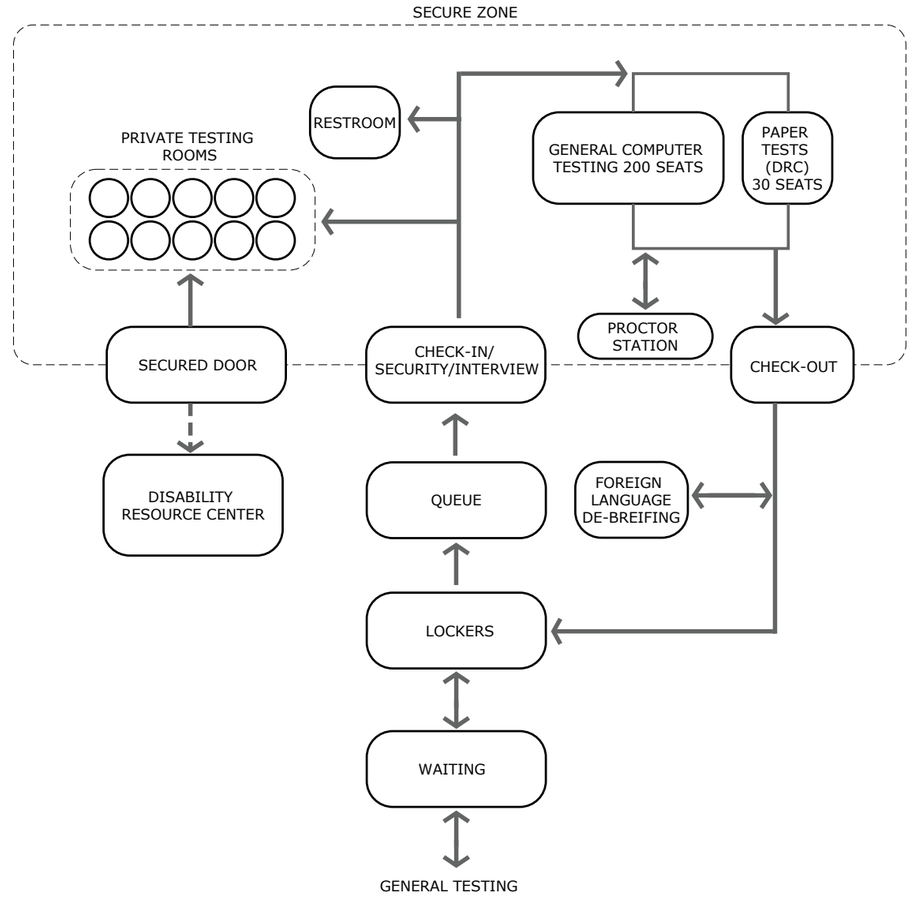
Central Testing Center - General Testing

1. All general testing students (including students taking accommodated testing through the DRC) enter the testing center through a waiting room/lounge space.
2. Students are instructed through signage to place all their items in lockers before checking in.
3. Students wait in line to check in.
4. Students check in and staff ensures compliance with the rules for their particular test (i.e. what items students can bring with them, or if they have to sit for an interview)
5. Students taking foreign language placement testing sit for the pre-test interview.
6. Students are assigned a testing station and directed toward the testing area appropriate for their test, or to a private testing room.
7. Proctors monitor tests live and via video monitor from the proctoring station.
8. Upon completion of the test, students check out.
9. Students taking foreign language testing go to the debriefing station to review their test results.
10. Students go to their lockers to retrieve their items.
11. Students exit through the waiting area.

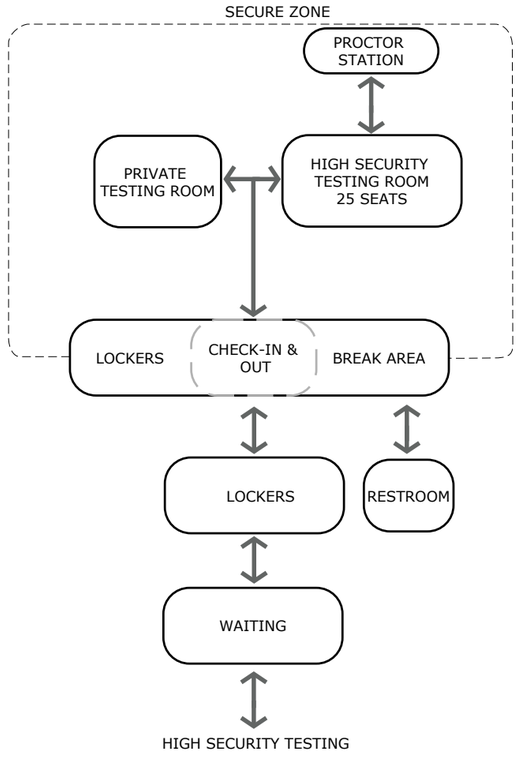
Central Testing Center – High-Security Testing

1. All high-security testing students (including students taking accommodated testing through the DRC) enter the testing center through a waiting area.
2. Students wait in line to check in and are assigned a locker.
3. If required by the test, students complete the pre-test requirements (i.e., reading a pamphlet)
4. Students place all their items in lockers.
5. Students go through the security procedures required for that test (signature pad, palm vein scanning, thumbprint, picture taken).
6. Students are individually walked to their testing station or private room.
7. During the course of the test, students may freely take a break or use the restroom in the secured break area. This area will be separated from the testing center via a one-way door. After taking a break, students will check back in and be walked to their testing station.
8. Proctors will monitor tests live and via video monitor from the proctoring station.
9. Upon completion of the test, students raise their hand and are escorted from their testing station.
10. Students will go to their lockers to retrieve their items.
11. Students will exit through the waiting area.

GENERAL TESTING CIRCULATION DIAGRAM



HIGH-SECURITY TESTING CIRCULATION DIAGRAM



HIGH-SECURITY TESTING CENTER REQUIREMENTS

A testing center must meet strict requirements to receive certification to administer a vendor's tests. The following requirements are a compilation of the requirements of testing vendors Pearson VUE, CLEP, and Prometric. In each case, the most stringent of each vendor's requirement has been compiled in an effort to meet the requirements of all vendors. It will be important for this information to be confirmed before proceeding with actual construction of the testing space. The individual requirements of each vendor can be found in the appendix of this document.

Staff

Based on 25 testing stations, a staff of two people is required to be physically on site to operate the center when in use. The total testing center staff must contain at least two certified TAs. One certified TA must be present in the testing center at all times.

Admissions Desk

This is the area where students first interact with a testing center representative. It also is where students check out during breaks and completion of the test.

Requirements

- Web camera
- Palm vein device
- Signature pad
- Hand-held metal detector
- Chair for candidate photographs, 2' from counter
- Area monitored by a camera with audio

Proctor/Surveillance Station

This workstation is where the test proctor will monitor testing in progress.

Requirements

- Viewing window into room
- Monitors to view video surveillance of testing in progress
- Telephone
- Area monitored by a camera with audio

Lockers

Students will store all their belongings in a locker during testing.

Requirements

- One locker per work station
- Must be outside the exam delivery room
- Key must be given to student

Testing Room

This is the physical space that houses the individual testing stations.

Requirements

- Individual test station screens positioned perpendicular to entrance so monitors cannot be seen from entrance
- Stations should not be at right angles to each other
- A Pearson VUE video camera should be centered above the partition between every other workstation
- A Prometric video camera should be centered above the partition between every other workstation
- No more than 25 test stations
- Must contain a wall clock, mounted to be visible to exam candidates
- Room must be entered directly from reception area with a lockable door
- Room must be lit using glare-free lighting
- A microphone must be centrally located to record room audio
- Palm scans are required at each entrance and exit of the room
- A white noise system should be designed into the space
- Surfaces should be finished in acoustically absorptive materials

Testing Stations

The station where test candidates will be seated to take tests.

Requirements

- Comfortable chair with back and height adjustment
- Test cannot be administered on a laptop
- Single 19-24" monitor, non-touchscreen, aspect ratio of 16:9 or 16:10. Note that Pearson VUE testing requires GPU scaling with this aspect ratio
- Four-foot-wide desk surface
- Partitions between stations, 5' high, projecting 4' from back of desk

Private Testing Room
Individual testing room where a single candidate can take a test in isolation, with space for an accommodation assistant when necessary.

Requirements
- Sound-isolated room
- Door with viewing window
- Two comfortable chairs with back and height adjustment
- Test cannot be administered on a laptop
- Single 19-24" monitor, non-touchscreen, aspect ratio of 16:9 or 16:10. Note that Pearson VUE testing requires GPU scaling with this aspect ratio
- Five-foot-wide desk surface

Server Room
The area to house the local server/cache proxy for storing test delivery information and test packages as well as temporary storage of candidate results.

Requirements
- Secure space
- Connected to all exam delivery workstations through a LAN
- Server should never be turned off
- Can be in a remote location
- Server cannot be a laptop

INSTRUCTIONAL DATA PROCESSING

The goal of Instructional Data Processing (IDP) is to support the educational process at Purdue University by providing fast, reliable, accurate test scoring and evaluation services while embracing the most current technologies. IDP manages course evaluations, test proctoring services, Scantron grading, specialized grade book work for certain departments, and Scantron scoring in support of research.

CURRENT FACILITIES AND CONDITIONS

Space Program
- Area for a high-capacity scanner (Scantron iNSIGHT 150)
  - Requires precise humidity control
  - Requires a dust-free environment
  - An acoustically isolated space is ideal as the scanner produces a high volume of noise
- Transaction counter for interacting with customers and taking Scantron sheets
- Storage area to keep one month's worth of Scantron sheets
- Secure office with a meeting space
- Desks for four staff members in an open area
- Incoming/outgoing Scantron test storage area

Space Requirements
- Independently secured department space
- The department should be in an easily accessible location for faculty but not in a high student traffic zone to lessen security concerns

Current Test Grading Load
- 400,000 to 500,000 tests graded per semester
- 75,000 to 85,000 tests graded during finals week

Current Personnel
- Four staff members and one director

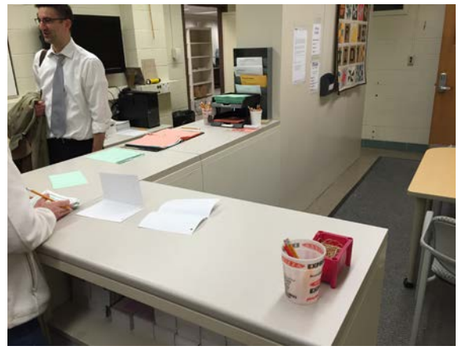
Hours of Operation
- Standard Hours
  - Monday, Tuesday, Thursday, Friday: 8 AM to 5 PM
  - Wednesday: 8 AM to 1 PM and 2 PM to 5 PM
- Final Exam Week Hours
  - Monday through Thursday: 8 AM to 8 PM
  - Friday: 8 AM to 10 PM
  - Saturday: 10 AM to 6 PM
  - Sunday: Noon to 2 PM

Current Conditions
Instructional Data Processing is located in the Steward Center in Room G-39. This location generally works well for the department as it is somewhat centrally located, yet not in an area of the building with a high amount of student traffic. Spatially, the department's area is somewhat oversized; however, the current space is not functioning well in several ways. Because the grading machine is not acoustically isolated, it makes adjacent spaces difficult to conduct meetings, and because the HVAC is not properly regulating humidity for the area, the grading machine has issues operating at maximum efficiency. Co-locating this department with the DRC would result in increased test-grading efficiency as the DRC is one of IDP's major clients. Professors with students utilizing the testing center in the DRC must currently travel to the DRC first to pick up completed tests to then combine with tests completed in their classroom before dropping them off at IDP to be graded. There may be another opportunity for efficient use of staff between the test proctoring services administered by IDP and the need for test proctors at the proposed centralized testing center.

Potential Areas for Improvement

- Need a six- to seven-person meeting space for proctor training
- Need a better conditioned and humidity controlled space to keep the grading machine running smoothly
- Need an acoustically isolated space to separate noise from the grading machine from the rest of the office
- Would like windows into their office
- A centralized Scantron pick-up/drop-off location near the front reception desk of the renovated Potter Hall may increase efficiency and ease of use.

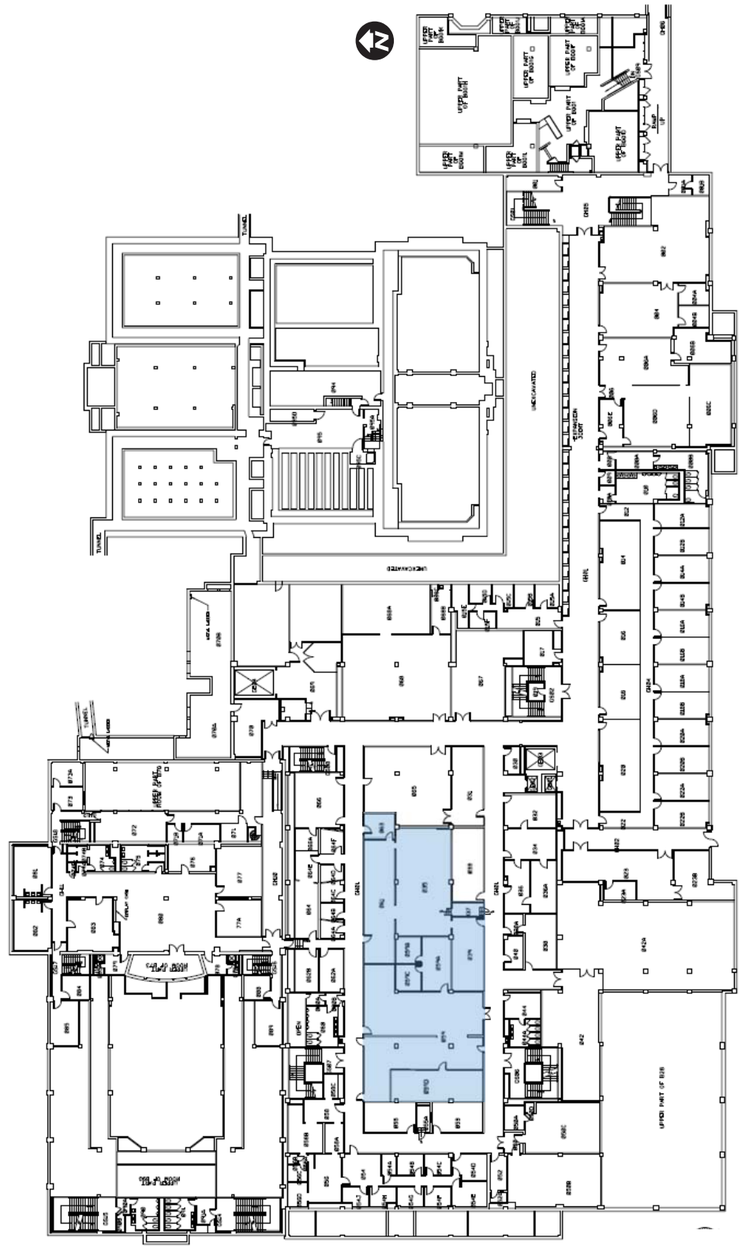
Current Photos







Current Floor Plan STEWART CENTER (STEW) - GROUND FLOOR



Space Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	4	41	164
Director	1	120	120
Check-In/Reception	1	100	100
Grand Total	**6**	**261**	**384**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Data Processing Equipment	1	200	200
Files/Storage	1	100	100
Reception	1	300	300
Grand Total	**3**	**600**	**600**
		TOTAL	**984**

DISABILITY RESOURCE CENTER

The Disability Resource Center (DRC) ensures qualified students with disabilities equal access to all University programs, services, and activities. DRC supports students with disabilities by coordinating reasonable and appropriate accommodations, educating the University community about disability issues and rights, and promoting an informed and inclusive campus environment that actively addresses individual and institutional barriers.

CURRENT FACILITIES AND CONDITIONS

Space Requirements

- Easily accessible first-floor location
- Panic button under all staff desks
- All desks, doorways, and furniture should be universally accessible, beyond ADA where applicable.
- Scooters/wheelchairs should be considered.
- Accommodation specialist offices should contain a separate meeting table for four and be acoustically isolated.
- Two to three ADA-accessible parking spaces should be adjacent to the building.

Current Conditions

The Disability Resource Center is currently located in Young Hall on the eighth floor in Room 830. The location on the eighth floor presents a challenge to the primary clients of this department, who are often students in wheelchairs/scooters. At present, Purdue is one of only two schools in the Big Ten that does not locate its DRC facility on the first floor. The DRC department is currently operating in limited space and could use more room. This will become an even larger issue as the department continues to grow over time.

Future Growth

The DRC at Purdue has seen a substantial growth in the number of students that self disclose and request reasonable accommodations. The most frequently utilized accommodation is accommodated testing. DRC data reflects 677 students were registered with the DRC for fall 2010. That number has grown to 1,256 students registered for services for fall 2015. This reflects an 85.5 percent growth over this time period.

A published article in the Journal of Vocational Rehabilitation reports that as a result of current legislation and labor market trends, the number of students with disabilities has risen to 17% students attending postsecondary education programs.[1] The results of the “American Community Survey” estimates the overall rate of disability in the US population in 2014 was 12.6%[2]. From the National Center for Educational Statistics Fast Facts sheet, data reflects for 2011-2012 the percentage distribution of disabled students is 11.1% of the student body.

If we consider Purdue’s total undergraduate enrollment to be at 39,000 students, it would appear the DRC’s presently reported numbers underrepresent disabled student numbers actually attending the university. As society continues to improve its understanding of disability as another form of diversity, civil rights, and social justice, we can expect students to overcome the societal stigma of disability. This will result in the continued increase of the number of students self-reporting.

Potential Areas for Improvement

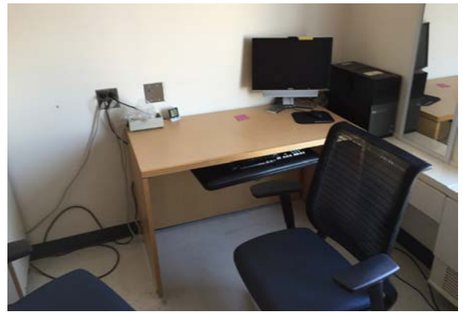
- Relocate department to a first-floor location
- Provide more room in the reception/waiting areas
- Greater access to daylight and a more welcoming environment
- Alternate Format Production area needs additional space

[1] Journal of Vocational Rehabilitation, vol. 16, no. 3,4, pp. 189-198, 200 by Stodden, Whelley, Chang, Harding reports, 42
[2] www.DisabilityCompendium.org

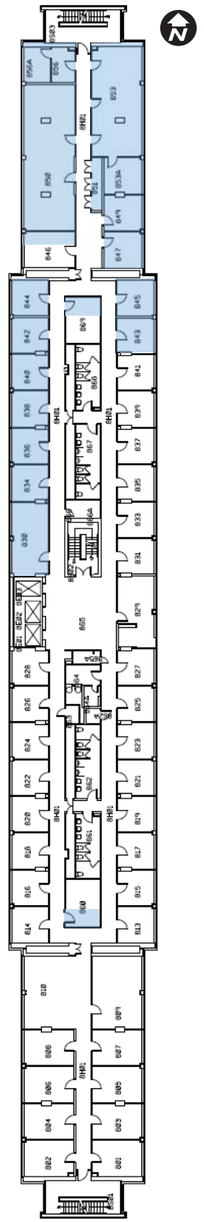
Current Photos








Current Floor Plan ERNEST C. YOUNG HALLL (YONG) - EIGHTH FLOOR



Space Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Assistant Director	2	120	240
Director	1	120	120
Undergrad Assistant	2	40	80
Accommodation Specialist	7	160	1,120
Grand Total	**14**	**481**	**1,642**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Alternative Format Production	1	350	350
Files/Storage	1	60	60
Reception	1	300	300
Work Café	1	200	200
Workroom	1	150	150
Grand Total	**5**	**1060**	**1,060**
		TOTAL	**2,702**

ASSISTIVE TECHNOLOGY CENTER

The Assistive Technology Center (ATC) promotes an accessible information technology infrastructure at Purdue that is usable by all students, faculty, and staff regardless of disability status. The ATC provides students with disabilities the accommodations to do schoolwork, in much the same way the DRC currently provides students with disabilities the accommodations to take tests.

CURRENT FACILITIES AND CONDITIONS

The ATC is located in Stewart Center Room 111. The space has recently been outfitted with new furniture, and the current size is sufficient for its function. Much like the DRC, it is a space regularly accessed by students with disabilities, so the space should be designed beyond code minimums and instead be designed with universal accessibility in mind.

Hours of Operation

- Monday through Thursday: 7 AM to 11 PM
- Friday: 7 AM to 11 PM
- Saturday: 9 AM to 6 PM
- Sunday: Noon to 11 PM

Space Program

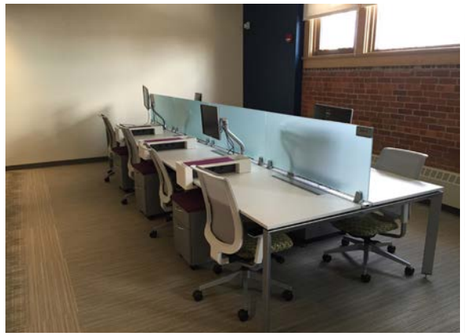
Position	Number of Personnel	Area Per Person	Total Area
Director	1	120	120
Grand Total	**1**	**120**	**120**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Assistive Technology Center	1	850	850
Grand Total	**1**	**850**	**850**
		TOTAL	**970**

COLLEGE OF ENGINEERING

In 2014, the College of Engineering at Purdue conducted a campus-wide facilities master plan that outlined the future plans for growth of the College in accordance with the College's Strategic Growth Initiatives. The plan identified the Potter Center as a home for the College's Student Resource Programs. The plan did not, however, include the new CTC, DRC, ATC, or IDP. The goal of this study will be to find synergies and efficiencies among these programs in order to co-locate these departments within the Potter Center while simultaneously respecting the goals of the College of Engineering.

In addition to housing the new departments, it is the goal of the College to create an efficient space suitable for 21st century education. Using Grissom Hall as a benchmark, the College would like to re-image the building, bringing closed office areas in toward the center and creating open office areas around the perimeter. This strategy will allow the maximum amount of daylight penetration into the building, and efficiency strategies such as "hotdesking" will ensure maximum utilization of the space.





RELOCATED COLLEGE OF ENGINEERING DEPARTMENTS

Environmental and Ecological Engineering (EEE)

The EEE program pursues a new and unique approach to environmental engineering. In addition to considering conventional waste emission control and treatment, students learn to use systems engineering techniques to proactively avoid environmental problems. This approach adopts an integrated perspective to ensure engineering systems are created that can exist in harmony with the natural world and serve society.

EEE builds upon the strong community of faculty and students engaged in environmentally related research at Purdue. EEE is an interdisciplinary program that is structured to be inclusive of all efforts across the College related to environmental and ecological engineering.

Space Program

Position	Number of Personnel	Area Per Person	Total Area
Academic Dean / Department Head	1	120	120
Admin/Operations Staff, Workstation	2	41	82
Clerical-Workstation	1	40	40
Grad-Thesis Desks (64 Students)	45	26	1170
Hoteling Offices	4	120	480
Management	2	120	240
Professional Staff - Office	3	120	360
Research - Visiting Faculty	4	120	480
Tenured/TT Faculty	10	120	1200
Visiting Scholars	4	64	256
Grand Total	**76**	**891**	**4,428**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Open Lab - Project Engineering	2	300	600
Reception	1	180	180
Student Organizations	1	120	120
Teaching Lab - Wet	1	1200	1,200
Teaching Lab - Wet/Dry	1	1200	1,200
Teaching Support - 25%	1	300	300
Mediascape Collaboration Space	1	150	150
Grand Total	**9**	**3570**	**3,870**
		TOTAL	**8,298**

Engineering Education (ENE)

The School of Engineering Education envisions a more inclusive socially connected and scholarly engineering education. This implies radically rethinking the boundaries of engineering and the purpose of engineering education. Their mission is to transform engineering education based on scholarship and research rests on three pillars: Reimagining engineering and engineering education, creating field-shaping knowledge, and empowering agents of change.

Space Program

Position	Number of Personnel	Area Per Person	Total Area
Adjuncts/Joint Appointments	2	120	240
Communications Director	1	120	120
Director of Project Assessment	1	120	120
Director, Development	1	120	120
Emeritus	2	120	240
Grad-Thesis Desks (160 Students)	112	26	2,912
Instructor/Lecturer	2	120	240
Managing Director	1	120	120
Post Doctoral Assistants - Workstation	9	40	360
Teaching Assistant	20	40	800
Tenured/TT Faculty	28	120	3,360
Visiting Scholars	4	64	256
Grand Total	**183**	**1,130**	**8,888**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Focus/TA Consulting/Help/Advising	14	100	1,400
Student Organizations	1	120	120
Student Societies	3	120	360
MS Distance Learning Classroom (Opt.)	1	1,440	1,440
Workroom	1	120	120
Grand Total	**20**	**1,900**	**3,440**
		TOTAL	**12,328**

Engineering Advising (ENE Adv.)

Housed within the Engineering Education program, First-Year Engineering has a staff of professional advisors, available to help students in a variety of academic success, registration, personal, and professional issues. The advisors are here to be a student's definitive source of information and advice; students can consult with their advisors any time they have a question or problem, no matter how small.

Global Engineering Program (GEP)

Purdue's Office of Global Engineering Programs firmly believes that immersive, transformative international experiences such as studying, working, or researching abroad lead to more effective engineers and graduates who can work effectively across cultural, ethnic, and national boundaries and help businesses succeed in an increasingly interconnected world.

The GEP program can play a key role in translating innovations to scalable solutions for eradicating extreme poverty, hunger, improving global health, building resilient societies, and empowering vulnerable populations. The program focuses on educating globally competent engineers and creating global impact.

Space Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	1	41	41
Assistant Director	1	120	120
Associate Director	2	120	240
Director	1	120	120
Grad-Thesis Desks (160 Students)	1	26	26
Management - Office Manager	1	120	120
Management - Staff	1	120	120
Program Manager	1	120	120
Undergrad Assistant	2	40	80
Grad Student Admin. Workstation (Hotdesk)	3	64	192
Grand Total	**14**	**891**	**1,179**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Storage	1	120	120
Student Ambassador Workroom	1	360	360
Workroom	1	120	120
Grand Total	**4**	**720**	**720**
		TOTAL	**1,899**

Office of Professional Practice (OPP)

The OPP facilitates the experiential education and professional practice of Purdue University students within the academic environment of the institution and its global partners, participates in academic research within the field of professional practice, and assists academic units with enhanced employer engagement. Committed to fostering a holistic student experience, OPP offers formal programs in 5-Session Co-Op, 3-Session Co-Op, internship, and research formats. OPP also facilitates the Global Engineering Alliance for Research and Education (GEARE), Purdue's flagship global engineering program. Purdue designs professional practice programs to combine practical on-the-job experiences with the classroom training of a four-year college curriculum

Space Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Assistant Director	3	120	360
Clerical-Workstation	3	40	120
Director	1	120	120
Grand Total	**9**	**321**	**682**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Workroom	1	120	120
Grand Total	**2**	**240**	**240**
		TOTAL	**922**

College of Engineering Honors (HONORS)

Honors at Purdue is designed to provide the highly motivated and academically successful student with a broader, more enriched educational experience during the first year of college through a variety of academic challenges. The program focuses on students looking for a higher degree of intensity in their undergraduate studies.

Space Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	4	41	164
Advisor	3	120	360
Assistant Director	1	120	120
Associate Director	1	120	120
Director	1	120	120
Grand Total	**10**	**521**	**884**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	360	360
Grand Total	**1**	**360**	**360**
		TOTAL	**1,244**

Minority Engineering Program (MEP)

Established in 1974, MEP has been nationally recognized and replicated to advance engineering learning, discovery, and engagement in fulfillment of the Land Grant promise through outreach, recruitment, and retention of historically underrepresented students in their pursuit to become extraordinary Purdue engineers. Although our programs are open to all students without regard to race, ethnicity, or gender, underrepresented students at Purdue University include African Americans, Latino/Hispanic Americans, Native Americans, and Hawaiians/Pacific Islanders.

Space Program

Position	Number of Personnel	Area Per Person	Total Area
Assistant Director	1	120	120
Associate Director	1	120	120
Data Analyst	1	120	120
Director	1	120	120
Grad Student Admin. Workstation (Hotdesk)	2	64	128
Grand Total	**6**	**544**	**608**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Tutorial Center	1	900	900
Workroom	1	120	120
Grand Total	**3**	**1,140**	**1,140**
		TOTAL	**1,748**

Office of Future Engineers (OFE)

OFE aspires to inspire middle school through high school students to consider engineering as a career and is dedicated to recruiting a diverse and academically gifted student body to the First-Year Engineering (FYE) program at Purdue University.

Space Program

Position	Number of Personnel	Area Per Person	Total Area
Associate Director	1	120	120
Director	1	120	120
Management - Office Manager	1	120	120
Management - Staff	1	120	120
Grad Assistant Workstation (Hotdesk)	2	64	128
Undergrad Staff Workstations (Hotdesk)	7	40	280
Grand Total	**13**	**584**	**888**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Workroom	1	120	120
Grand Total	**2**	**240**	**240**
		TOTAL	**1,128**

Women in Engineering Program (WIEP)
From mentoring to career development, WIEP continually encourages current and future women engineering students by providing interesting and engaging programming relevant to their lives. WIEP programs are a place to learn, discover, and explore aspects of engineering and connect with others who are also interested. They are a place of encouragement, support, and positive perspectives for those who are interested in following their dreams by pursuing an engineering degree.

Space Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	1	41	41
Associate Director	1	120	120
Communications	1	120	120
Director	1	120	120
Management	1	120	120
Professional Staff - Office	2	120	240
Program Manager	1	120	120
Research - Visiting Faculty	1	120	120
Grad Assistant Workstation (Hotdesk)	6	64	384
Grad Student Admin. Workstation (Hotdesk)	6	64	384
Grand Total	**21**	**1,009**	**1,769**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Student Organizations	1	120	120
Workroom	1	120	120
Grand Total	**3**	**360**	**360**
		TOTAL	**2,129**

FOOD SERVICE PROGRAM

The food program at Potter Center is intended to provide an element of hospitality to a building designed to provide a unique experience for prospective students, current students, and the faculty and staff of the engineering program at Purdue University.

Potter Center will also accommodate non-engineering students using the general testing facility. As the master plan for this area and the transformation of Potter Center each becomes a reality, there will be a definite increase in pedestrian traffic directly in front of this building.

Providing food service at this location follows a trend on university campuses nationwide. There is an expectation among students that food service, in varying degrees and in different formats, will be accessible and available throughout the day and across campus. No longer are traditional dining halls or even updated food courts the only location for food venues on campus.

At Potter Center, we envision three distinct retail concepts on the first floor. As currently depicted on the plan, the food venues will be adjacent to a café/co-working space and in close proximity to a significant number of staff and faculty offices. The food service area is close enough to the street to allow access to pedestrian traffic. It is also in the path of students headed to the lower level for testing purposes.

The support kitchen allows for shared preparation, storage, and warewashing facilities adjacent to the retail spaces. There will also be additional food storage available in the basement. This type of arrangement is currently in place and functioning on the Purdue University campus at Memorial Union and at Indiana University at the Bookmarket Eatery (located in the basement of Wells library and the result of a 2016 renovation). With the advent of food halls in the commercial food service sector, the use of shared space for back-of-house functions has become even more commonplace.

For the purpose of this exercise, we have not specifically identified the three venues that will operate at Potter Center. However, based the trends we are observing at the collegiate level among the Gen Z population, we are safe to make a few assumptions about the offerings here. The role food will play in this environment is one of hospitality and convenience. With Potter Center being a gateway into the engineering school, the entry spaces need to reflect that. With that in mind, a coffee café venue will fit well into the food service program at this location. It should be positioned to allow for varying hours of operation from the other two venues.

Each of the other concepts will need to be unique from other offerings in this area of campus and flexible enough to perhaps change as trends change or even just have enough adaptability to allow for different menus for different dayparts. The physical design of the space can allow for this to great degree based on the equipment package selected for each concept.

The setup for the food venues will also allow for limited preparation and cooking to be done in front of the customer. This will promote an environment of freshness and allow the patron some control over their exact food selection. This food presentation and serving strategy mimics that of the increasingly popular “fast casual” restaurants. Students (Gen Z) are very comfortable with this service style based on where they dine when they are not on campus and even when they are in other areas of this campus, as well. “Fresh” and “customizable” are two key attractants for retail food environments today.

A renovated Potter Center will be designed to serve a diverse student population within a flagship academic program for Purdue University. While the food venues may not be the reason for a lot of the traffic in and around this building, it will certainly play a major role in enhancing the student and staff experience in this facility.

PHASING

Throughout the course of this study, there have been myriad discussions regarding phasing the renovations to the building. Phasing the departments that have the greatest need first, such as the DRC and CTC, is discussed in part one of this study. Here, we'll address adding dining to the building as "phase 1."

The following floor plan shows the first floor of the existing Potter building in grayscale, with the dining functions from the Option D study overlaid in color. As you can see from the plan, the dining areas could conceivably be created at the center of the building, and the dining areas could be created along the south side. The south wall of the proposed dining areas are among the only existing glassy areas in Potter, so this dining space fortunately lands in the perfect location for this use.

Until the rest of the project is completed, the areas surrounding the dining area will obviously be awkwardly shaped, and there is certain to be a loss of efficiency in the building. One awkward-shaped space on this floor could conceivably function as a temporary cold storage area (ideally the triangular space just north of the kitchen). This space could be shifted to its permanent home on the lower level in phase two of the renovations.

Also of note is the expanded restroom. Due to the increase in occupant load, these restrooms will need to be enlarged to be code compliant and ADA compliant. No restrooms within Potter are currently ADA accessible. This restroom is in close enough proximity to the existing restroom that plumbing renovations should be manageable.

We would recommend the elevator, the renovation of the balance of the restrooms, the reskinning of the facades, and the infill of the mezzanine occur during the second phase of the project. However, for dining to be the most successful, it will need to be visible and welcoming to students, so we recommended the site area south of Potter receive a complete renovation from the south face of Potter south to Centennial Mall during the dining renovations. This renovation will create the exterior dining and gathering areas that will become an integral part of the dining experience. Additionally, we have proposed a new vestibule be created along the southern side of the building demarcating a clear entrance point to the dining area.

However, while the dining appears to function well on the first floor, the renovation will need to address other issues on other floors of the building. Of particular concern are the chases that will be necessary to exhaust the kitchen areas. Kitchen exhaust chases are large, and running this large chase through the second and third floor area will not only be a challenge, but will certainly be inconvenient to the occupants of the upper floors.

We recommend the second phase of the renovation encompass the balance of the project. As discussed in the previous phase of the study, attempting to put some of the functions in early, especially the DRC and CTC would require those functions to shut down and be relocated during construction. The dining function would also likely need to be shut down during phase two of the renovations, especially if the renovation involves reskinning the façade.


1S02
289 SF
136C
207 SF
127B
142 SF
1H03
192 SF
136
196 SF
136B
105 SF
127
1,708 SF
136A
138 SF
127A
142 SF
138
138 SF
122
553 SF
1H04
238 SF
143
24 SF
123
24 SF
146
236 SF
1E01
80 SF
1S03
183 SF
131
94 SF
119
48 SF
118
605 SF
135
1,082 SF
117
137 SF
Storage
123 SF
Cooler
141 SF
Freezer
97 SF
135C
273 SF
114
628 SF
Kitchen
642 SF
Support
140 SF
Venue 3
145 SF
114A
114 SF
Dishwashing
209 SF
Food Service Circ
689 SF
165
Redundant Room
1E02
39 SF
Jan
88 SF
Venue 2
185 SF
1S04
120 SF
Women
206 SF
Men
200 SF
Venue 1
176 SF
114C
95 SF
114B
157 SF
175
24 SF
Corridor
888 SF
102
732 SF
Elec
24 SF
Lobby
788 SF
1S05
124 SF
Engineering Cafe
1,688 SF
1H01
2,303 SF
1H02
174 SF
158
102 SF
Vestibule
127 SF
1S01
290 SF
157
141 SF
151
84 SF
152
69 SF
153
69 SF
154
117 SF
155
127 SF

OPTION E – DRC, ATC, CTC, COLLEGE OF ENGINEERING PROGRAMS, ENGINEERING IN EDUCATION, AND FOOD SERVICE

This option builds on the goals for Option D while simultaneously integrating food service into the building.

Programmatically, the building was essentially "filled" in option D; therefore, spatial tradeoffs are necessary from the Option D design to accommodate the food service functions. However, given the clear synchronicity between the goal of the COE programs to introduce students to the field of engineering and the inherent draw of food service bringing students into the building, the tradeoffs may well be worth the reduction in space for the increase in exposure. If designed effectively, there should be opportunities for both planned and impromptu exposure of the COE's departments. Planned exposure could be through planned displays, screens, or installations of student work. Impromptu exposure could come from carefully selected and well integrated furniture and technology design that encourages group work and collaboration between current engineering students to take place within dining spaces also used by non-engineering students.

It is worth mentioning that at the time of this study, the program being used for all departments was created nearly two years ago in the early months of 2016, prior to the construction of the ALC, and likely many other projects on campus. If the recommendations of this study are implemented, we strongly recommend the current programmatic needs be reassessed to truly evaluate the impact of the space changes, not only between Option D and E but also relative to the current needs of all the impacted programs.

GOALS

The goals of Option E are to:

- create a more accessible and functional Disability Resource Center.
- increase efficiency and effectiveness by co-locating the DRC and the ATC.
- increase efficiency, cost control, academic integrity, and ease of use by creating a centralized location housing all campus testing needs and combining all campus testing functions in one location.
- create a new campus wide resource that is appropriately sized and staffed to deliver better testing conditions at Purdue with increased academic integrity.
- reduce the demand for academic and non-academic space during periods of high testing volume.
- realize the College of Engineering's master plan for the Potter Center by creating a center devoted to serving first-year engineering students and associated recruitment and retention programs.
- locate ENE alongside FYE and COE programs to realize the natural synergies between these departments and buildings.
- update the interior and exterior of the Potter Center and fully integrate the building into the surrounding campus.
- create a new dining venue to serve students in the COE as well as those in the ALC and passers by using centennial mall.

ARCHITECTURAL DESIGN

Option E plays a careful balancing act among the needs of all programs contained in the building. Because the goal of many of the included programs are aimed at increasing access, accessibility, and exposure to students, space on the first floor is in high demand. With the exception of the testing

center, an argument could be made for nearly every other program to be housed on the first floor. However, because first floor space is limited, we have included only the programs with the greatest need for exposure or accessibility on this level. These functions are the DRC, which has an obvious need for at-grade access; food service, which will greatly benefit from maximized exposure; and the COE's programs, which focus on recruitment of students to the College of Engineering.

In contrast to the Option D plan, the Option E first floor plan trades the vast majority of the communal work cafes and a significant number of conference and focus rooms for food service and dining areas. An argument could be made that these are an even trade since the dining areas could easily function as meeting spaces and gathering areas, especially if thoughtfully designed.

The tutorial center has been shifted from the first floor to the lower level, adjacent to the testing center. It is likely the center will benefit by being located on a quieter floor, but it may also suffer from lack of exposure. The Data Processing department has similarly been shifted down to the lower level. This department does not require (or want) exposure to students, but it would benefit from easy access by faculty. The lower level could be argued to be slightly harder to access since it requires changing floors; however, the existing data processing center is also in a basement location and functions well there. The 100-person conference room, the new "Fu Room," has been shifted to the second level. Due to its size, it would be more conveniently located on the first floor to keep 100 people from needing to travel upstairs. Because a conference room does not need to "attract" users, the food service function won first-floor priority.

The elimination and shifting of these spaces then accommodates the food service function. In food service, material flow is important, beginning with deliveries, through storage, then preparation, and finally delivery to the customer. Potter's loading dock, service entrance, and existing freight elevator are ideally suited to this use. Deliveries from the dock can either pass directly into the back of house kitchen area or down the elevator into the cold storage area. Neither route conflicts with the regular movements of students or faculty in the building.

The addition of dining to the first floor, with the subsequent increase in occupants, creates a new requirement on this floor—increasing the quantity of restroom fixtures. This further decreases the amount of otherwise assignable area at ground level and increases the cost of this option.

It is always the goal of the design team to provide offices with windows, or at least indirect access to daylight through borrowed lights. However, due to the quantity of first floor offices and Potter's large square footprint that allows little daylight to the core, there are just over a dozen offices on this floor without access to daylight. This is in contrast to the Option D plan, where every closed office had at least limited daylight access. This could be fixed by shifting these offices to the third floor, where there is currently unused space and locations for daylit offices. However, this shift would separate related programs and potentially reduce opportunities for synchronicities between programs. This issue warrants further exploration with the College of Engineering.

On the lower level, the testing center remains nearly identical to the Option D plan. North of the testing center, the impact of the food service functions can be seen in the dramatic reduction of square footage given to the i2i Design/Project Studio and elimination of the i2i classroom. It was the sense of the University that these functions may be redundant to functions in the ALC and could be eliminated or reduced. Cold storage to serve the food service areas are located as near as possible to the elevator. Data processing and the tutorial center round out this floor's functions.

As in Option D, the second floor still accommodates the entire Engineering Education Department. However, new in this option is that it also houses the 100 person conference room, the Fu Room, and an associated pre-function area immediately adjacent the elevator. To accommodate this space, the second i2i/Honors Teaching Lab formerly on this floor has been eliminated. A reconfigured second floor plan has a slightly increased number of conference, focus, and work rooms in an effort to balance out the reduction of these spaces on the first floor.

INTERIOR DESIGN

Corridors will have new terrazzo, with vinyl base flexibility for future renovation. The existing terrazzo is both difficult to match and expensive to replace. In the long term, new terrazzo is extremely cost-effective and durable.

The entrance/commons will be a combination of terrazzo and carpet tile/vinyl, epoxy-painted walls, and accents of wallcovering or digital wallcovering. Administration areas will receive carpet tile, vinyl base, and wallcovering. All offices, workstations, grads, classrooms/computer labs, and conference rooms will have carpet tile and wall paint with accent colors. Carpet tile, rather than broadloom, is easy to replace if a tile is damaged. Break rooms/lounges/coffee bar/commons will have luxury vinyl tile (LVT), which is durable and easy to maintain. LVT comes in multiple colors and patterns that can give anything from a warm to industrial appearance. Teaching labs and workrooms will have sealed concrete floors, epoxy painted walls, and exposed ceilings.

Stairwells and other existing walls with exposed aggregate finishes will receive a hard plaster coating. To negate any unevenness in the surface, the plaster will receive a slight texture and paint. This will create a long-lasting durable surface that can easily be applied over the existing wall finish.

STRUCTURAL DESIGN

As a result of the reimaging of the building, an elevator will be added within the existing floor plan. At the location of the elevator, an opening will be required at each of the floor framing levels and at the roof framing level. The floor framing may require reinforcing around the opening to support the design floor/roof load. At the basement level, an opening will be required in the slab-on-grade to accommodate an elevator pit. The elevator pit construction will consist of reinforced cast-in-place concrete walls supported on a reinforced concrete mat slab that makes up the bottom of the elevator pit. The elevator shaft itself can either be framed with light-gauge metal studs and gypsum board sheathing or framed with full-height reinforced concrete masonry units. One advantage to framing with reinforced concrete masonry units would be the possibility of using the elevator shaft walls to assist in supporting the floor and roof systems that had holes cut in them to accommodate the elevator construction.

CIVIL DESIGN

Option E proposes to renovate all of the plaza space south of the Potter Center to serve two goals. First, it will provide an easy, directly accessible route to Potter Center's main entry. More than any other building on campus, Potter Center will be used by wheelchair-bound and disabled students. Creating an entry that allows these students to walk directly toward the door with their peers is a critical aspect of the success of the DRC within Potter. Additionally, renovation of the existing plaza can create more welcoming, inviting, and useable space outside Potter. In its current configuration, the building itself is obscured by very large trees and the patio area is difficult to access. The proposed configuration of Centennial Mall only addresses the Wilmeth Active Learning Center and makes no gestures of inclusion toward Potter. The revised design pulls from the distinctive geometry of Potter's curved staircase, radiating out through the plaza and creating a welcoming avenue toward the main entry. A terraced outdoor classroom creates a semi-private outdoor seating area, an interesting option for outdoor teaching, and serves as a subtle reminder that this is a place of learning.

MECHANICAL, ELECTRICAL AND PLUMBING DESIGN

Mechanical Systems, HVAC

- Repurposing the building not only justifies new mechanical systems but will require systems that meet the current energy code.
- The new basement floor plan shows approximately 50% of the assignable space is occupied by the testing centers while the remainder of the assignable area is made up of fairly typical office/academic spaces.
- The first floor is made up typical office/academic spaces except for the kitchen, food service area and the dining area.
- The second floor is all typical office/academic space.

- The third floor is mostly office/academic space except for the two wet teaching labs and the open space lab.
- Ideally, each floor would have its own air handling system. This better allows for changes to the building floor plan over the life of the mechanical system.
- Of the four floors, the basement is the most problematic. About 50% of the assignable floor space is dedicated to the testing centers. The design population density is extremely high but there will be times when there is essentially no thermal or ventilation load in the space. This tends towards localized sensible only cooling with a dedicated outside air system. However, the remainder of the floor tends towards a more traditional VAV reheat system. The deciding factor on which systems to use will depend on the anticipated frequency of usage of the testing centers and the resulting life cycle cost analysis based on first cost and operating cost.
- The first floor, even with the kitchen, lends itself to a single air handler system. It is not unusual for a kitchen with a grease hood and a dishwasher to have a make-up air unit, but in this case, there may not be room available in the basement mechanical room. If a make-up air unit is used, it will probably have to sit on the roof and require additional chase space through the second and third floors. A roof mounted unit could also be problematic because in addition to the grease hood and dishwasher exhaust, there will also be chemical fume hood exhaust from the wet labs, all of which require separation from the make-up air unit. A single air handling unit, properly controlled, can serve the kitchen as well as the remainder of the floor area. One further note, the proposed floor plan shows three serving venues. It is possible one or more of the venues will require an exhaust hood, depending on what type of food service the venues support.
- Of the four floors, the second floor is the most uniform and can be served by a single air handler located in the penthouse.
- The nature of the wet labs on the third floor has yet to be defined. If the labs have a low fume hood density, serving the entire floor with a single air handler located in the penthouse is a reasonable approach. Even if the labs have a high fume hood density, it is likely the exhaust requirements will not over-power a single air handler system. It has been shown in the Brown Hall teaching labs, fume hood sashes are open to working height approximately one hour out of a three-hour lab, when operated correctly. With the labs being a small percentage of the total area, at this time a separate system for the labs cannot be justified.
- The new elevator will be equipped with a hoist way vent and normally open vent damper controlled in accordance with the code. Measures will also be taken to control odors associated with hydraulic elevators.
- While no new mechanical room space is anticipated, it is anticipated additional chase space will be required to run larger, more efficient ductwork.

Mechanical Systems, Plumbing

- The existing utilities to the building should be adequate for the proposed building usage, but all systems need to be evaluated for capacity once the usage is better defined.
- The age of the domestic water heater is not known. If it is found to be the original water heater or if it has significant aging, it can be replaced with a more efficient steam-to-water heater that sub cools the steam condensate.
- Additions/modifications to the existing plumbing system will be made as required to accommodate floor plan changes. The kitchen on the first floor will require a grease interceptor be placed on grade.

Fire Protection Systems

- It is assumed the existing incoming service is adequate and no changes are required.
- It is anticipated the existing fire mains can remain in place and new laterals run as required. If a main is found to be in conflict with a requirement of the new floor plan, the main will be rerouted or offset for the new work.

Electrical Systems

- Replace 208/120V fused-switch switchboard (M-L-1 & M-P-1) with 2500A breaker-style switchboard.
- Replace 480/2770V fused-switch switchboard(MSB-1) with 800A breaker-style switchboard.
- Provide Arc reduction breaker in separate room between secondary of existing transformers (Trans L1, Trans P1 & Trans P2 and new switchboards (M-P-1, M-P-2 & MSB-1).
- Maintain four existing bus ducts that run vertically in the northwest, northeast, southwest and southeast electrical room.
- Replace existing 400A fused-style distribution boards and 200A branch circuit panelboard(s) in each of the four electrical rooms with new 400 A circuit breaker type distribution board and 200A circuit breaker type panelboards. Provide additional panelboard(s) to serve the increased load in Kitchen/Cafeteria and Testing Area.
- Remove all receptacles and lights in renovated area. Associated conduit and conductors will be removed back to panelboard.
- Remove all branch circuit/feeder (conduit and conductors) serving HVAC, plumbing, and other equipment to be demolished.
- Install new lighting throughout the building and new lighting control to comply with 2010 Energy code. New branch circuits from new panelboards in each of the four electrical rooms.
- Install new power receptacles throughout the building. New branch circuits from new panelboards in each of the four electrical rooms.
- Provide circuits to new HVAC and plumbing equipment as required.
- MCC-G-1 and MCC-PH will be replace depending on the final determined needs of the building systems.

Telecom System Changes

- Remove all existing CAT 5e cabling from entire building, coordinating timing so that critical equipment does not go down.
- Add PICS as required for reimaging of entire building
- Install CAT 6A cabling throughout building to all PICS and wireless access points.
- Install new CAT6A patch panels in existing BDF and IDF rooms. Equipment racks will need to have 6" vertical wire management on the sides. They will not require horizontal cable management.
- Establish new minimum 8x8' IDF rooms on first, second, and third floors, ideally stacked.
- Install new telecom grounding system throughout building.
- Install new 4" skeletal raceway and basket cable tray in renovated areas as required.
- Demo all existing ECN cabling out of building and back into utility tunnel.
- Install voice over IP (VOIP) telephone system.
- Add a redundant 24-strand single-mode fiber fed from a second campus telecom node in Brown Hall. Purdue ITaP would install cable into the building, but the cost would be absorbed by this project.
- Demo the existing 250-pair OSP copper cable and replace with a 25-pair into the building.
- Purdue ITaP would install cable into the building, but the cost would be absorbed by this project.

Fire Alarm System

- Redesign of complete system, but main fire alarm panel could be retained.
- Monitoring for kitchen hood suppression systems as required.

SPACE PROGRAM

Disability Resource Center

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Assistant Director	2	120	240
Director	1	120	120
Undergrad Assistant	2	40	80
Accommodation Specialist	7	160	1,120
Total	**14**	**481**	**1,642**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Alternative Format Production	1	350	350
Files/Storage	1	60	60
Reception	1	300	300
Work Café	1	200	200
Workroom	1	150	150
Total	**5**	**1060**	**1,060**
		TOTAL	**2,702**

Assistive Technology Center

Position	Number of Personnel	Area Per Person	Total Area
Director	1	120	120
Total	**1**	**120**	**120**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Assistive Technology Center	1	850	850
Total	**1**	**850**	**850**
		TOTAL	**970**

Potter Center Business Unit

Position	Number of Personnel	Area Per Person	Total Area
Business Manager	1	120	120
Business Staff Workstation	6	41	246
Grand Total	**7**	**161**	**366**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	80	80
Grand Total	**1**	**80**	**80**
		TOTAL	**446**

Central Testing Center

Position	Number of Personnel	Area Per Person	Total Area
HS Testing - Office	1	120	120
HS Testing - Video Proctor	1	90	90
HS Testing - Work Stations	4	90	360
Online Compliance - Private Office	1	120	120
Online Compliance - Shared office	3	120	360
Testing - Administration Counter	1	90	90
Testing - Director	1	120	120
Testing - Office	2	120	240
Testing - Video Proctor	1	90	90
Grand Total	**15**	**960**	**1,590**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
General Testing Stations	22.5	230	5,175
High Security Testing Stations	26	25	650
Lockers, 250	300	1	300
Lockers, 35	30	1	30
Private Testing Room	48	21	1,008
Queueing for 60	540	1	540
Secure Break Area	150	1	150
Secure Restrooms	64	6	384
Secure Storage	200	1	200
Storage	50	1	50
Testing Check-in	400	2	800
Testing Concierge	200	1	200
Grand Total	**2,030.5**	**291**	**9,487**
		TOTAL	**11,077**

Community Resources

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Focus/TA Consulting/Help/Advising	16	120	1,920
Reception	1	250	250
Conference Room - 12 People	6	300	1,800
Conference Room - 20 People	2	470	940
Conference Room - 6 People	8	200	1,600
Work Café	4	500	2,000
Conference Room - 100 People	1	1,000	1,000
Grand Total	**38**	**2,840**	**9,510**
		TOTAL	**9,510**

Instructional Data Processing

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	4	41	164
Director	1	120	120
Check-In/Reception	1	100	100
Grand Total	**6**	**261**	**384**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Data Processing Equipment	1	200	200
Files/Storage	1	100	100
Reception	1	300	300
Grand Total	**3**	**600**	**600**
		TOTAL	**984**

First-Year Engineering (FYE)

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Advisor	10	120	1,200
Instructional Support Coordinators	3	120	360
Instructional Support Manager	1	120	120
Instructional Support Specialist	1	120	120
Assistant Head	1	120	120
Student Advisor Workstations (Shared)	4	40	160
Total	**22**	**681**	**2,162**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	200	200
Teaching Lab Support	1	200	200
i2i Classroom	1	2,800	2,800
i2i Honors Teaching Lab ENGR 141/142	1	3,000	3,000
i2i Design / Project Studio	1	3,000	3,000
Total	**5**	**9,200**	**9,200**
		TOTAL	**11,362**

Office of Future Engineers

Position	Number of Personnel	Area Per Person	Total Area
Associate Director	1	120	120
Director	1	120	120
Management - Office Manager	1	120	120
Management - Staff	1	120	120
Grad Assistant Workstation (Hotdesk)	2	64	128
Undergrad Staff Workstations (Hotdesk)	7	40	280
Grand Total	**13**	**584**	**888**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Workroom	1	120	120
Grand Total	**2**	**240**	**240**
		TOTAL	**1,128**

Office of Professional Practice

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Assistant Director	3	120	360
Clerical-Workstation	3	40	120
Director	1	120	120
Grand Total	**9**	**321**	**682**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Workroom	1	120	120
Grand Total	**2**	**240**	**240**
		TOTAL	**922**

College of Engineering Education (ENE)

Position	Number of Personnel	Area Per Person	Total Area
Adjuncts/Joint Appointments	2	120	240
Communications Director	1	120	120
Director of Project Assessment	1	120	120
Director, Development	1	120	120
Emeritus	2	120	240
Grad-Thesis Desks (160 Students)	112	26	2,912
Instructor/Lecturer	2	120	240
Managing Director	1	120	120
Post Doctoral Assistants - Workstation	9	40	360
Teaching Assistant	20	40	800
Tenured/TT Faculty	28	120	3,360
Visiting Scholars	4	64	256
Grand Total	**183**	**1,130**	**8,888**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Focus/TA Consulting/Help/Advising	14	100	1,400
Student Organizations	1	120	120
Student Societies	3	120	360
MS Distance Learning Classroom (Opt.)	1	1,440	1,440
Workroom	1	120	120
Grand Total	**20**	**1,900**	**3,440**
		TOTAL	**12,328**

Women In Engineering Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	1	41	41
Associate Director	1	120	120
Communications	1	120	120
Director	1	120	120
Management	1	120	120
Professional Staff - Office	2	120	240
Program Manager	1	120	120
Research - Visiting Faculty	1	120	120
Grad Assistant Workstation (Hotdesk)	6	64	384
Grad Student Admin. Workstation (Hotdesk)	6	64	384
Grand Total	**21**	**1,009**	**1,769**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Student Organizations	1	120	120
Workroom	1	120	120
Grand Total	**3**	**360**	**360**
		TOTAL	**2,129**

College of Engineering Honors

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	4	41	164
Advisor	3	120	360
Assistant Director	1	120	120
Associate Director	1	120	120
Director	1	120	120
Grand Total	**10**	**521**	**884**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	360	360
Grand Total	**1**	**360**	**360**
		TOTAL	**1,244**

Global Engineering Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	1	41	41
Assistant Director	1	120	120
Associate Director	2	120	240
Director	1	120	120
Grad-Thesis Desks (160 Students)	1	26	26
Management - Office Manager	1	120	120
Management - Staff	1	120	120
Program Manager	1	120	120
Undergrad Assistant	2	40	80
Grad Student Admin. Workstation (Hotdesk)	3	64	192
Grand Total	**14**	**891**	**1,179**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Storage	1	120	120
Student Ambassador Workroom	1	360	360
Workroom	1	120	120
Grand Total	**4**	**720**	**720**
		TOTAL	**1,899**

Minority Engineering Program

Position	Number of Personnel	Area Per Person	Total Area
Assistant Director	1	120	120
Associate Director	1	120	120
Data Analyst	1	120	120
Director	1	120	120
Grad Student Admin. Workstation (Hotdesk)	2	64	128
Grand Total	**6**	**544**	**608**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Tutorial Center	1	900	900
Workroom	1	120	120
Grand Total	**3**	**1,140**	**1,140**
		TOTAL	**1,748**

Division of Environmental and Ecological Engineering (EEE)

Position	Number of Personnel	Area Per Person	Total Area
Academic Dean / Department Head	1	120	120
Admin/Operations Staff, Workstation	2	41	82
Clerical-Workstation	1	40	40
Grad-Thesis Desks (64 Students)	45	26	1170
Hoteling Offices	4	120	480
Management	2	120	240
Professional Staff - Office	3	120	360
Research - Visiting Faculty	4	120	480
Tenured/TT Faculty	10	120	1200
Visiting Scholars	4	64	256
Grand Total	**76**	**891**	**4,428**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Open Lab - Project Engineering	2	300	600
Reception	1	180	180
Student Organizations	1	120	120
Teaching Lab - Wet	1	1200	1,200
Teaching Lab - Wet/Dry	1	1200	1,200
Teaching Support - 25%	1	300	300
Mediascape Collaboration Space	1	150	150
Grand Total	**9**	**3570**	**3,870**
		TOTAL	**8,298**

Net Area Total, All Departments	66,747
Grossing Factor	30%
NET AREA, ALL DEPARTMENTS	**86,771**

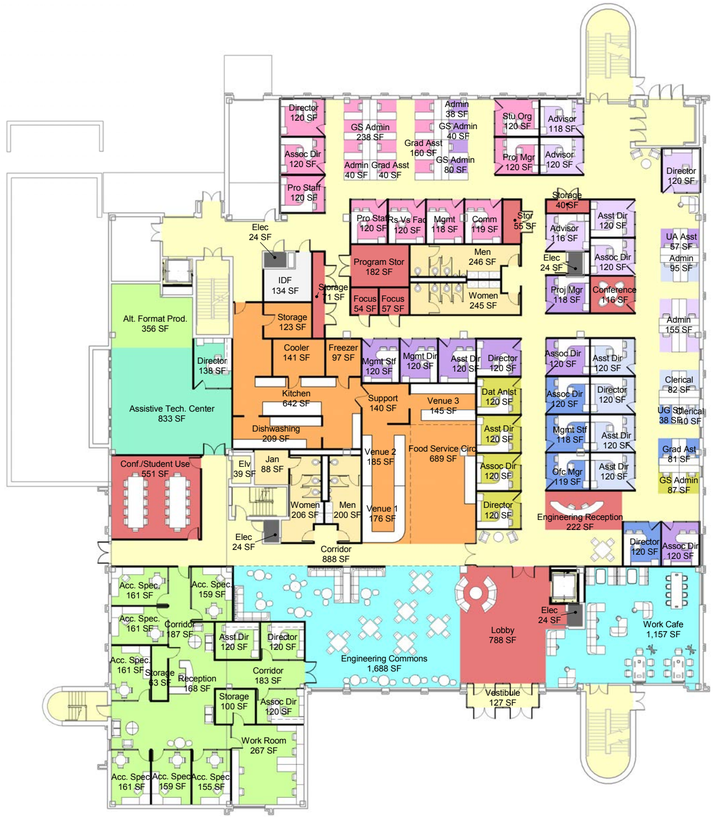
DISPLACED EXISTING AREA

Department	Area Per Department
COE Performance Space	541
College of Engineering	2,149
Corporate and Foundation Relations	700
Electrical and Computer Engineering	5,295
Engineering	2,863
Engineering Computer Network	711
Engineering Research Center	7,909
Environmental and Ecological Engineering	3,865
General Building	2,132
IEEE	479
Industrial Engineering	3,795
IT Cust Rltn	202
Laboratory of Renewable Resources Engineering	10,265
Libraries	23,365
Mechanical Engineering	703
Nuclear Engineering	9,817
Operations and Maintenance	2,943
Grand Total	**93,866**



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COMMUNITY RESOURCE
- DINING
- DISABILITY RESOURCE CENTER
- ENGINEERING CAFE / COWORKING AREA
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION E
LOWER LEVEL PLAN



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COMMUNITY RESOURCE
- DINING
- DISABILITY RESOURCE CENTER
- ENGINEERING CAFE / COWORKING AREA
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION E
FIRST FLOOR PLAN



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COMMUNITY RESOURCE
- DINING
- DISABILITY RESOURCE CENTER
- ENGINEERING CAFE / COWORKING AREA
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION E
SECOND FLOOR PLAN



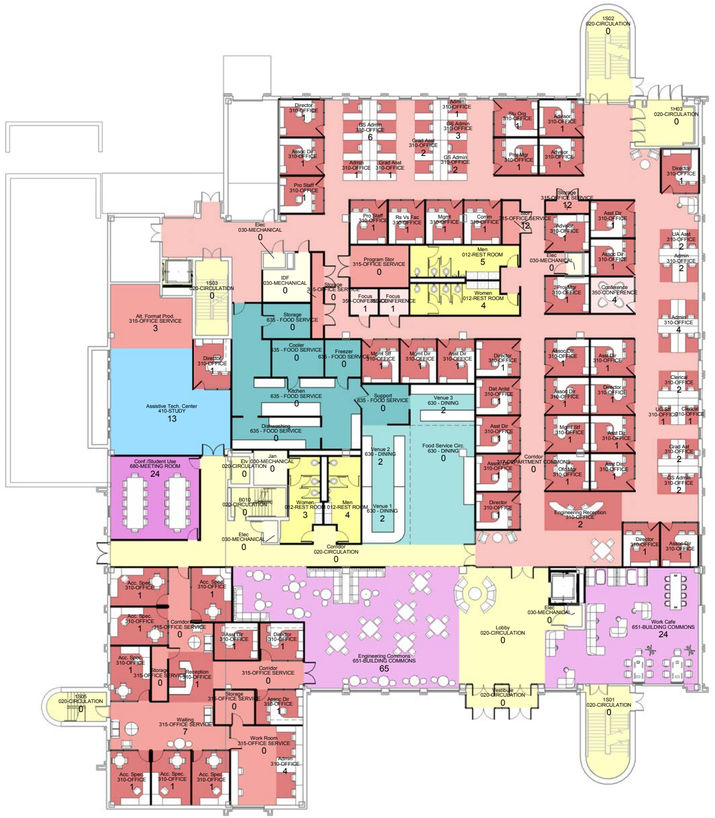
- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COMMUNITY RESOURCE
- DINING
- DISABILITY RESOURCE CENTER
- ENGINEERING CAFE / COWORKING AREA
- ENGINEERING EDUCATION
- ENGINEERING HONORS
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- FIRST YEAR ENGINEERING
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- GENERAL BUILDING - MECHANICAL
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- OPEN / FLEX
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION E
THIRD FLOOR PLAN



- 010-BUILDING SERVICE
- 012-REST ROOM
- 020-CIRCULATION
- 030-MECHANICAL
- 210-CLASS LAB
- 250-NONCLASS LAB
- 255-NONCLASS LAB SERVICE
- 310-OFFICE
- 315-OFFICE SERVICE
- 317-DEPARTMENT COMMONS
- 350-CONFERENCE
- 410-STUDY
- 455-STUDY SERVICE
- 630 - DINING
- 635 - FOOD SERVICE
- 650-LOUNGE
- 651-BUILDING COMMONS
- 680-MEETING ROOM
- 685 - MEETING ROOM SERVICE
- 730-CENTRAL STORAGE
- Calculating...

OPTION E
LOWER LEVEL USE



- 010-BUILDING SERVICE
- 012-REST ROOM
- 020-CIRCULATION
- 030-MECHANICAL
- 210-CLASS LAB
- 250-NONCLASS LAB
- 255-NONCLASS LAB SERVICE
- 310-OFFICE
- 315-OFFICE SERVICE
- 317-DEPARTMENT COMMONS
- 350-CONFERENCE
- 410-STUDY
- 455-STUDY SERVICE
- 630 - DINING
- 635 - FOOD SERVICE
- 650-LOUNGE
- 651-BUILDING COMMONS
- 680-MEETING ROOM
- 685 - MEETING ROOM SERVICE
- 730-CENTRAL STORAGE
- Calculating...

OPTION E
FIRST FLOOR USE


Stair
020-CIRCULATION
0
Elec
030-MECHANICAL
0
Elev
030-MECHANICAL
0
IDF
030-MECHANICAL
0
Elec
030-MECHANICAL
0
Elec
030-MECHANICAL
0
Elev
030-MECHANICAL
0
Elec
030-MECHANICAL
0
Elec
030-MECHANICAL
0
Elec



- 010-BUILDING SERVICE
- 012-REST ROOM
- 020-CIRCULATION
- 030-MECHANICAL
- 210-CLASS LAB
- 250-NONCLASS LAB
- 255-NONCLASS LAB SERVICE
- 310-OFFICE
- 315-OFFICE SERVICE
- 317-DEPARTMENT COMMONS
- 350-CONFERENCE
- 410-STUDY
- 455-STUDY SERVICE
- 630 - DINING
- 635 - FOOD SERVICE
- 650-LOUNGE
- 651-BUILDING COMMONS
- 680-MEETING ROOM
- 685 - MEETING ROOM SERVICE
- 730-CENTRAL STORAGE
- Calculating...

OPTION E
THIRD FLOOR USE

Conceptual Rendering of Engineering Commons - Option E



Conceptual Aerial of South Facade and Plaza - Option E



Conceptual South Facade and Plaza - Option E



Conceptual Plaza Redesign - Option E



OPTION F

OPTION F – DRC, ATC, CTC, COLLEGE OF ENGINEERING PROGRAMS, ENGINEERING IN EDUCATION, AND FOOD SERVICE

Option F is nearly identical to Option E, with one exception. The proposed elevator location has been shifted outside the existing building footprint, creating a functional and unified circulation path and adding a highlight feature to the façade.

GOALS

The goals of Option E are to:

- create a more accessible and functional Disability Resource Center.
- increase efficiency and effectiveness by co-locating the DRC and the ATC.
- increase efficiency, cost control, academic integrity, and ease of use by creating a centralized location housing all campus testing needs and combining all campus testing functions in one location.
- create a new campus wide resource that is appropriately sized and staffed to deliver better testing conditions at Purdue with increased academic integrity.
- reduce the demand for academic and non-academic space during periods of high testing volume.
- realize the College of Engineering's master plan for the Potter Center by creating a center devoted to serving First-Year Engineering students and associated recruitment and retention programs.
- locate ENE alongside FYE and COE Programs to realize the natural synergies between these departments and buildings.
- update the interior and exterior of the Potter Center and fully integrate the building into the surrounding campus.
- create a new dining venue to serve students in the COE as well as those in the ALC and passersby using centennial mall.
- create a unified circulation path throughout the renovated building while adding a highlight façade feature.

ARCHITECTURAL DESIGN

In all other options, the elevator has not been ideally located on every floor. The new elevator location is primarily driven by the testing center, which naturally can't have an elevator open directly into the testing area. Therefore, it must be located along the hallway between the two existing and code-required means of egress. Since a new main entrance and reception area was created on the south side of the building on the first floor, the elevator then naturally ends up located at the south end of the building. On floors two and three, however, this location proves somewhat awkward.

Option D solves this design issue by properly locating the elevator near the existing stair shaft, creating a cohesive and functional circulation core. This location easily serves all the building's floors without breaking up the floor plan or creating indirect circulation routes.

An additional aesthetic benefit comes from this location as well. Removing and replacing the current stair creates an opportunity to add a feature element onto the building. If thoughtfully designed, this element could speak to the engineering disciplines contained within the redesigned Potter Center and give a new degree of life to the exterior of the building as students visibly move up and down the stairway.

While the relocation of the stair benefits all the floors, the greatest benefit can be seen on the first floor. The Option D reception and dining space was effectively split into two halves, and the view into the College of Engineering's reception area was somewhat hidden by the elevator. In Option E, this is no longer the case. The COE's reception area is clearly visible, and the space has been unified into a contiguous area, creating a greater potential flexibility in its uses, clearer and simpler circulation, and greater daylight penetration.

INTERIOR DESIGN

Corridors will have new terrazzo, with vinyl base flexibility for future renovation. The existing terrazzo is both difficult to match and expensive to replace. In the long term, new terrazzo is extremely cost-effective and durable.

The entrance/commons will be a combination of terrazzo and carpet tile/vinyl, epoxy-painted walls, and accents of wallcovering or digital wallcovering. Administration areas will receive carpet tile, vinyl base, and wallcovering. All offices, workstations, grads, classrooms/computer labs, and conference rooms will have carpet tile and wall paint with accent colors. Carpet tile, rather than broadloom, is easy to replace if a tile is damaged. Break rooms/lounges/coffee bar/commons will have luxury vinyl tile (LVT), which is durable and easy to maintain. LVT comes in multiple colors and patterns that can give anything from a warm to industrial appearance. Teaching labs and workrooms will have sealed concrete floors, epoxy painted walls, and exposed ceilings.

Stairwells and other existing walls with exposed aggregate finishes will receive a hard plaster coating. To negate any unevenness in the surface, the plaster will receive a slight texture and paint. This will create a long-lasting durable surface that can easily be applied over the existing wall finish.

STRUCTURAL DESIGN

As a result of the reimaging of the building, an elevator will be added within the existing floor plan. At the location of the elevator, an opening will be required at each of the floor framing levels and at the roof framing level. The floor framing may require reinforcing around the opening to support the design floor/roof load. At the basement level, an opening will be required in the slab-on-grade to accommodate an elevator pit. The elevator pit construction will consist of reinforced cast-in-place concrete walls supported on a reinforced concrete mat slab that makes up the bottom of the elevator pit. The elevator shaft itself can either be framed with light-gauge metal studs and gypsum board sheathing or framed with full-height reinforced concrete masonry units. One advantage to framing with reinforced concrete masonry units would be the possibility of using the elevator shaft walls to assist in supporting the floor and roof systems that had holes cut in them to accommodate the elevator construction.

CIVIL DESIGN

Option F proposes to renovate all of the plaza space south of the Potter Center to serve two goals. First, it will provide an easy, directly accessible route to Potter Center's main entry. More than any other building on campus, Potter Center will be used by wheelchair-bound and disabled students. Creating an entry that allows these students to walk directly toward the door with their peers is a critical aspect of the success of the DRC within Potter. Additionally, renovation of the existing plaza can create more welcoming, inviting, and useable space outside Potter. In its current configuration, the building itself is obscured by very large trees and the patio area is difficult to access. The proposed configuration of Centennial Mall only addresses the Wilmeth Active Learning Center and makes no gestures of inclusion toward Potter. The revised design pulls from the distinctive geometry of Potter's curved staircase, radiating out through the plaza and creating a welcoming avenue toward the main entry. A terraced outdoor classroom creates a semi-private outdoor seating area, an interesting option for outdoor teaching, and serves as a subtle reminder that this is a place of learning.

MECHANICAL, ELECTRICAL AND PLUMBING DESIGN

Mechanical Systems, HVAC

- The Mechanical HVAC requirements for Option F below are the same as for Option D except as noted below for the new elevator which is to be added to the exterior of the building.
- Repurposing the building not only justifies new mechanical systems but will require systems that meet the current energy code.
- The new basement floor plan shows approximately 50% of the assignable space is occupied by the testing centers while the remainder of the assignable area is made up of fairly typical office/academic spaces.
- The first floor is made up typical office/academic spaces except for the kitchen, food service area and the dining area.
- The second floor is all typical office/academic space.
- The third floor is mostly office/academic space except for the two wet teaching labs and the open space lab.
- Ideally, each floor would have its own air handling system. This better allows for changes to the building floor plan over the life of the mechanical system.
- Of the four floors, the basement is the most problematic. About 50% of the assignable floor space is dedicated to the testing centers. The design population density is extremely high but there will be times when there is essentially no thermal or ventilation load in the space. This tends towards localized sensible only cooling with a dedicated outside air system. However, the remainder of the floor tends towards a more traditional VAV reheat system. The deciding factor on which systems to use will depend on the anticipated frequency of usage of the testing centers and the resulting life cycle cost analysis based on first cost and operating cost.
- The first floor, even with the kitchen, lends itself to a single air handler system. It is not unusual for a kitchen with a grease hood and a dishwasher to have a make-up air unit, but in this case, there may not be room available in the basement mechanical room. If a make-up air unit is used, it will probably have to sit on the roof and require additional chase space through the second and third floors. A roof mounted unit could also be problematic because in addition to the grease hood and dishwasher exhaust, there will also be chemical fume hood exhaust from the wet labs, all of which require separation from the make-up air unit. A single air handling unit, properly controlled, can serve the kitchen as well as the remainder of the floor area. One further note, the proposed floor plan shows three serving venues. It is possible one or more of the venues will require an exhaust hood, depending on what type of food service the venues support.
- Of the four floors, the second floor is the most uniform and can be served by a single air handler located in the penthouse.
- The nature of the wet labs on the third floor has yet to be defined. If the labs have a low fume hood density, serving the entire floor with a single air handler located in the penthouse is a reasonable approach. Even if the labs have a high fume hood density, it is likely the exhaust requirements will not over-power a single air handler system. It has been shown in the Brown Hall teaching labs, fume hood sashes are open to working height approximately one hour out of a three-hour lab, when operated correctly. With the labs being a small percentage of the total area, at this time a separate system for the labs cannot be justified.
- The new elevator will be equipped with a hoist way vent and normally open vent damper controlled in accordance with the code. The Owner has reported that some of the new elevator hoist ways mounted to the exterior of the building have experience condensation problems. This issue will be addressed in a manner that meets the code. Measures will also be taken to control odors associated with hydraulic elevators.
- While no new mechanical room space is anticipated, it is anticipated additional chase space will be required to run larger, more efficient ductwork.

Mechanical Systems, Plumbing

- The Mechanical plumbing requirements for Option F below are the same as for Option D.
- The existing utilities to the building should be adequate for the proposed building usage, but all systems need to be evaluated for capacity once the usage is better defined.
- The age of the domestic water heater is not known. If it is found to be the original water heater or if it has significant aging, it can be replaced with a more efficient steam-to-water heater that sub cools the steam condensate.

- Additions/modifications to the existing plumbing system will be made as required to accommodate floor plan changes. The kitchen on the first floor will require a grease interceptor be placed on grade.

Fire Protection Systems

- The Mechanical Fire Protection requirements for Option E below are the same as for Option D.
- It is assumed the existing incoming service is adequate and no changes are required.
- It is anticipated the existing fire mains can remain in place and new laterals run as required. If a main is found to be in conflict with a requirement of the new floor plan, the main will be rerouted or offset for the new work.

Electrical Systems

- Replace 208/120V fused-switch switchboard (M-L-1 & M-P-1) with 2500A breaker-style switchboard.
- Replace 480/2770V fused-switch switchboard(MSB-1) with 800A breaker-style switchboard.
- Provide Arc reduction breaker in separate room between secondary of existing transformers (Trans L1, Trans P1 & Trans P2 and new switchboards (M-P-1, M-P-2 & MSB-1).
- Maintain three existing bus ducts that run vertically in the northwest, northeast, and southwest electrical rooms.
- Remove existing busduct and associated panelboard(s) and feeder from southeast electrical room and replace with new distribution board and feeder from main electrical room to ground floor and second floor new southeast electrical room. Provide new panelboards on each floor from ground floor and second floor distribution boards.
- Replace existing 400A fused-style distribution boards and 200A branch circuit panelboard(s) in each of the three electrical rooms with new 400 A circuit breaker type distribution board and 200A circuit breaker type panelboards. Provide additional panelboard(s) to serve the increased load in Kitchen/Cafeteria and Testing Area.
- Remove all receptacles and lights in renovated area. Associated conduit and conductors will be removed back to panelboard.
- Remove all branch circuit/feeder (conduit and conductors) serving HVAC, plumbing, and other equipment to be demolished.
- Install new lighting throughout the building and new lighting control to comply with 2010 Energy code. New branch circuits from new panelboards in each of the four electrical rooms.
- Install new power receptacles throughout the building. New branch circuits from new panelboards in each of the four electrical rooms.
- Provide circuits to new HVAC and plumbing equipment as required.
- MCC-G-1 and MCC-PH will be replace depending on the final determined needs of the building systems.

Telecom System Changes

- Remove all existing CAT 5e cabling from entire building, coordinating timing so that critical
- equipment does not go down.
- Add PICS as required for reimaging of entire building
- Install CAT 6A cabling throughout building to all PICS and wireless access points.
- Install new CAT6A patch panels in existing BDF and IDF rooms. Equipment racks will need to have 6" vertical wire management on the sides. They will not require horizontal cable management.
- Establish new minimum 8x8' IDF rooms on first, second, and third floors, ideally stacked.
- Install new telecom grounding system throughout building.
- Install new 4" skeletal raceway and basket cable tray in renovated areas as required.
- Demo all existing ECN cabling out of building and back into utility tunnel.
- Install voice over IP (VOIP) telephone system.

- Add a redundant 24-strand single-mode fiber fed from a second campus telecom node in Brown Hall. Purdue ITaP would install cable into the building, but the cost would be absorbed by this project.
- Demo the existing 250-pair OSP copper cable and replace with a 25-pair into the building.
- Purdue ITaP would install cable into the building, but the cost would be absorbed by this project.

Fire Alarm System

- Redesign of complete system, but main fire alarm panel could be retained.
- Monitoring for kitchen hood suppression systems as required.

SPACE PROGRAM

Disability Resource Center

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Assistant Director	2	120	240
Director	1	120	120
Undergrad Assistant	2	40	80
Accommodation Specialist	7	160	1,120
Total	**14**	**481**	**1,642**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Alternative Format Production	1	350	350
Files/Storage	1	60	60
Reception	1	300	300
Work Café	1	200	200
Workroom	1	150	150
Total	**5**	**1060**	**1,060**
		TOTAL	**2,702**

Assistive Technology Center

Position	Number of Personnel	Area Per Person	Total Area
Director	1	120	120
Total	**1**	**120**	**120**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Assistive Technology Center	1	850	850
Total	**1**	**850**	**850**
		TOTAL	**970**

Potter Center Business Unit

Position	Number of Personnel	Area Per Person	Total Area
Business Manager	1	120	120
Business Staff Workstation	6	41	246
Grand Total	**7**	**161**	**366**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	80	80
Grand Total	**1**	**80**	**80**
		TOTAL	**446**

Central Testing Center

Position	Number of Personnel	Area Per Person	Total Area
HS Testing - Office	1	120	120
HS Testing - Video Proctor	1	90	90
HS Testing - Work Stations	4	90	360
Online Compliance - Private Office	1	120	120
Online Compliance - Shared office	3	120	360
Testing - Administration Counter	1	90	90
Testing - Director	1	120	120
Testing - Office	2	120	240
Testing - Video Proctor	1	90	90
Grand Total	**15**	**960**	**1,590**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
General Testing Stations	22.5	230	5,175
High Security Testing Stations	26	25	650
Lockers, 250	300	1	300
Lockers, 35	30	1	30
Private Testing Room	48	21	1,008
Queueing for 60	540	1	540
Secure Break Area	150	1	150
Secure Restrooms	64	6	384
Secure Storage	200	1	200
Storage	50	1	50
Testing Check-in	400	2	800
Testing Concierge	200	1	200
Grand Total	**2,030.5**	**291**	**9,487**
		TOTAL	**11,077**

Community Resources

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Focus/TA Consulting/Help/Advising	16	120	1,920
Reception	1	250	250
Conference Room - 12 People	6	300	1,800
Conference Room - 20 People	2	470	940
Conference Room - 6 People	8	200	1,600
Work Café	4	500	2,000
Conference Room - 100 People	1	1,000	1,000
Grand Total	**38**	**2,840**	**9,510**
		TOTAL	**9,510**

Instructional Data Processing

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	4	41	164
Director	1	120	120
Check-In/Reception	1	100	100
Grand Total	**6**	**261**	**384**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Data Processing Equipment	1	200	200
Files/Storage	1	100	100
Reception	1	300	300
Grand Total	**3**	**600**	**600**
		TOTAL	**984**

First-Year Engineering (FYE)

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Advisor	10	120	1,200
Instructional Support Coordinators	3	120	360
Instructional Support Manager	1	120	120
Instructional Support Specialist	1	120	120
Assistant Head	1	120	120
Student Advisor Workstations (Shared)	4	40	160
Total	**22**	**681**	**2,162**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	200	200
Teaching Lab Support	1	200	200
i2i Classroom	1	2,800	2,800
i2i Honors Teaching Lab ENGR 141/142	1	3,000	3,000
i2i Design / Project Studio	1	3,000	3,000
Total	**5**	**9,200**	**9,200**
		TOTAL	**11,362**

Office of Future Engineers

Position	Number of Personnel	Area Per Person	Total Area
Associate Director	1	120	120
Director	1	120	120
Management - Office Manager	1	120	120
Management - Staff	1	120	120
Grad Assistant Workstation (Hotdesk)	2	64	128
Undergrad Staff Workstations (Hotdesk)	7	40	280
Grand Total	**13**	**584**	**888**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Workroom	1	120	120
Grand Total	**2**	**240**	**240**
		TOTAL	**1,128**

Office of Professional Practice

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Assistant Director	3	120	360
Clerical-Workstation	3	40	120
Director	1	120	120
Grand Total	**9**	**321**	**682**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Workroom	1	120	120
Grand Total	**2**	**240**	**240**
		TOTAL	**922**

College of Engineering Education (ENE)

Position	Number of Personnel	Area Per Person	Total Area
Adjuncts/Joint Appointments	2	120	240
Communications Director	1	120	120
Director of Project Assessment	1	120	120
Director, Development	1	120	120
Emeritus	2	120	240
Grad-Thesis Desks (160 Students)	112	26	2,912
Instructor/Lecturer	2	120	240
Managing Director	1	120	120
Post Doctoral Assistants - Workstation	9	40	360
Teaching Assistant	20	40	800
Tenured/TT Faculty	28	120	3,360
Visiting Scholars	4	64	256
Grand Total	**183**	**1,130**	**8,888**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Focus/TA Consulting/Help/Advising	14	100	1,400
Student Organizations	1	120	120
Student Societies	3	120	360
MS Distance Learning Classroom (Opt.)	1	1,440	1,440
Workroom	1	120	120
Grand Total	**20**	**1,900**	**3,440**
		TOTAL	**12,328**

Women In Engineering Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	1	41	41
Associate Director	1	120	120
Communications	1	120	120
Director	1	120	120
Management	1	120	120
Professional Staff - Office	2	120	240
Program Manager	1	120	120
Research - Visiting Faculty	1	120	120
Grad Assistant Workstation (Hotdesk)	6	64	384
Grad Student Admin. Workstation (Hotdesk)	6	64	384
Grand Total	**21**	**1,009**	**1,769**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Student Organizations	1	120	120
Workroom	1	120	120
Grand Total	**3**	**360**	**360**
		TOTAL	**2,129**

College of Engineering Honors

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	4	41	164
Advisor	3	120	360
Assistant Director	1	120	120
Associate Director	1	120	120
Director	1	120	120
Grand Total	**10**	**521**	**884**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	360	360
Grand Total	**1**	**360**	**360**
		TOTAL	**1,244**

Global Engineering Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	1	41	41
Assistant Director	1	120	120
Associate Director	2	120	240
Director	1	120	120
Grad-Thesis Desks (160 Students)	1	26	26
Management - Office Manager	1	120	120
Management - Staff	1	120	120
Program Manager	1	120	120
Undergrad Assistant	2	40	80
Grad Student Admin. Workstation (Hotdesk)	3	64	192
Grand Total	**14**	**891**	**1,179**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Storage	1	120	120
Student Ambassador Workroom	1	360	360
Workroom	1	120	120
Grand Total	**4**	**720**	**720**
		TOTAL	**1,899**

Minority Engineering Program

Position	Number of Personnel	Area Per Person	Total Area
Assistant Director	1	120	120
Associate Director	1	120	120
Data Analyst	1	120	120
Director	1	120	120
Grad Student Admin. Workstation (Hotdesk)	2	64	128
Grand Total	**6**	**544**	**608**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Tutorial Center	1	900	900
Workroom	1	120	120
Grand Total	**3**	**1,140**	**1,140**
		TOTAL	**1,748**

Division of Environmental and Ecological Engineering (EEE)

Position	Number of Personnel	Area Per Person	Total Area
Academic Dean / Department Head	1	120	120
Admin/Operations Staff, Workstation	2	41	82
Clerical-Workstation	1	40	40
Grad-Thesis Desks (64 Students)	45	26	1170
Hoteling Offices	4	120	480
Management	2	120	240
Professional Staff - Office	3	120	360
Research - Visiting Faculty	4	120	480
Tenured/TT Faculty	10	120	1200
Visiting Scholars	4	64	256
Grand Total	**76**	**891**	**4,428**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Open Lab - Project Engineering	2	300	600
Reception	1	180	180
Student Organizations	1	120	120
Teaching Lab - Wet	1	1200	1,200
Teaching Lab - Wet/Dry	1	1200	1,200
Teaching Support - 25%	1	300	300
Mediascape Collaboration Space	1	150	150
Grand Total	**9**	**3570**	**3,870**
		TOTAL	**8,298**

Net Area Total, All Departments	66,747
Grossing Factor	30%
NET AREA, ALL DEPARTMENTS	**86,771**

DISPLACED EXISTING AREA

Department	Area Per Department
COE Performance Space	541
College of Engineering	2,149
Corporate and Foundation Relations	700
Electrical and Computer Engineering	5,295
Engineering	2,863
Engineering Computer Network	711
Engineering Research Center	7,909
Environmental and Ecological Engineering	3,865
General Building	2,132
IEEE	479
Industrial Engineering	3,795
IT Cust Rltn	202
Laboratory of Renewable Resources Engineering	10,265
Libraries	23,365
Mechanical Engineering	703
Nuclear Engineering	9,817
Operations and Maintenance	2,943
Grand Total	**93,866**



- ASSISTIVE TECHNOLOGY CENTER 0 SF of 994 SF
- BUSINESS OFFICE 0 SF of 653 SF
- CENTRAL TESTING CENTER 11,914 SF of 11,914 SF
- CIRCULATION 3,706 SF of 22,429 SF
- COMMUNITY RESOURCE 935 SF of 12,008 SF
- DINING 1,003 SF of 3,608 SF
- DISABILITY RESOURCE CENTER 0 SF of 3,834 SF
- ENGINEERING CAFE / COWORKING AREA 0 SF of 3,627 SF
- ENGINEERING EDUCATION 0 SF of 11,622 SF
- ENGINEERING HONORS 1,733 SF of 2,796 SF
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING 0 SF of 8,867 SF
- FIRST YEAR ENGINEERING 643 SF of 2,268 SF
- GENERAL BUILDING 63 SF of 151 SF
- GENERAL BUILDING - ELECTRICAL 1,760 SF of 2,048 SF
- GENERAL BUILDING - MECHANICAL 3,194 SF of 3,194 SF
- GENERAL BUILDING - PLUMBING 39 SF of 39 SF
- GENERAL BUILDING - TELECOMMUNICATIONS 280 SF of 751 SF
- GLOBAL ENGINEERING PROGRAM 0 SF of 1,211 SF
- INSTRUCTIONAL DATA PROCESSING 1,168 SF of 1,168 SF
- MINORITY ENGINEERING PROGRAM 966 SF of 1,676 SF
- OFFICE OF FUTURE ENGINEERS 0 SF of 751 SF
- ONLINE COMPLIANCE 538 SF of 538 SF
- OPEN / FLEX 0 SF of 6,276 SF
- PROFESSIONAL PRACTICE 0 SF of 823 SF
- RESTROOM 388 SF of 2,704 SF
- WOMEN IN ENGINEERING 0 SF of 1,725 SF

OPTION F
LOWER LEVEL PLAN



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COMMUNITY RESOURCE
- DINING
- DISABILITY RESOURCE CENTER
- ENGINEERING CAFE / COWORKING AREA
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION F
FIRST FLOOR PLAN



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COMMUNITY RESOURCE
- DINING
- DISABILITY RESOURCE CENTER
- ENGINEERING CAFE / COWORKING AREA
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION F
SECOND FLOOR PLAN



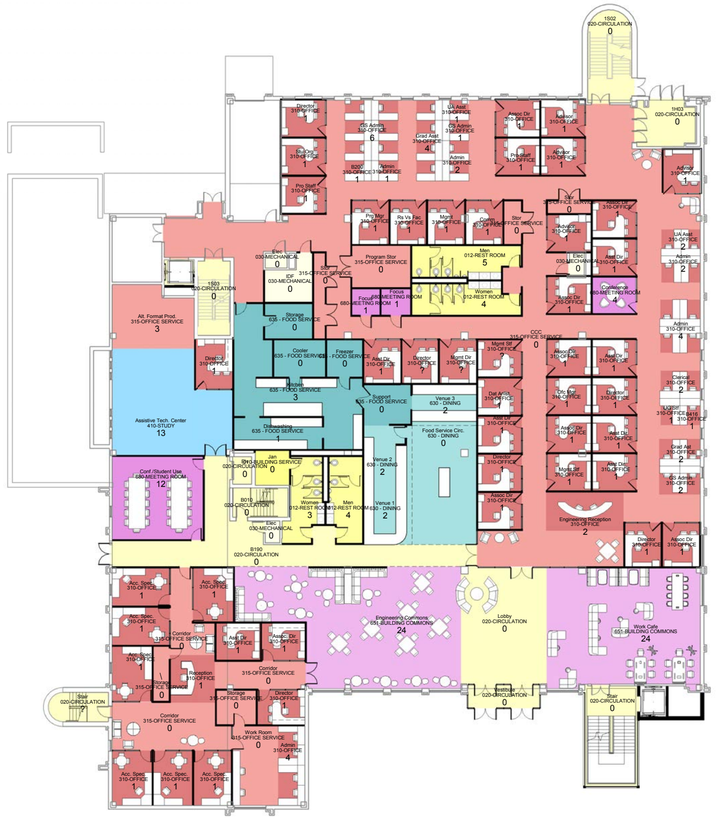
- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COMMUNITY RESOURCE
- DINING
- DISABILITY RESOURCE CENTER
- ENGINEERING CAFE / COWORKING AREA
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
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- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION F
THIRD FLOOR PLAN



- 010-BUILDING SERVICE
- 012-REST ROOM
- 020-CIRCULATION
- 030-MECHANICAL
- 210-CLASS LAB
- 250-NONCLASS LAB
- 255-NONCLASS LAB SERVICE
- 310-OFFICE
- 315-OFFICE SERVICE
- 317-DEPARTMENT COMMONS
- 350-CONFERENCE
- 410-STUDY
- 455-STUDY SERVICE
- 630 - DINING
- 635 - FOOD SERVICE
- 650-LOUNGE
- 651-BUILDING COMMONS
- 680-MEETING ROOM
- 685 - MEETING ROOM SERVICE
- 730-CENTRAL STORAGE
- Calculating...

OPTION F
LOWER LEVEL USE



- 010-BUILDING SERVICE
- 012-REST ROOM
- 020-CIRCULATION
- 030-MECHANICAL
- 210-CLASS LAB
- 250-NONCLASS LAB
- 255-NONCLASS LAB SERVICE
- 310-OFFICE
- 315-OFFICE SERVICE
- 317-DEPARTMENT COMMONS
- 350-CONFERENCE
- 410-STUDY
- 455-STUDY SERVICE
- 630 - DINING
- 635 - FOOD SERVICE
- 650-LOUNGE
- 651-BUILDING COMMONS
- 680-MEETING ROOM
- 685 - MEETING ROOM SERVICE
- 730-CENTRAL STORAGE
- Calculating...

OPTION F
FIRST FLOOR USE



- 010-BUILDING SERVICE
- 012-REST ROOM
- 020-CIRCULATION
- 030-MECHANICAL
- 210-CLASS LAB
- 250-NONCLASS LAB
- 255-NONCLASS LAB SERVICE
- 310-OFFICE
- 315-OFFICE SERVICE
- 317-DEPARTMENT COMMONS
- 350-CONFERENCE
- 410-STUDY
- 455-STUDY SERVICE
- 630 - DINING
- 635 - FOOD SERVICE
- 650-LOUNGE
- 651-BUILDING COMMONS
- 680-MEETING ROOM
- 685 - MEETING ROOM SERVICE
- 730-CENTRAL STORAGE
- Calculating...

OPTION F
SECOND FLOOR USE



- 010-BUILDING SERVICE
- 012-REST ROOM
- 020-CIRCULATION
- 030-MECHANICAL
- 210-CLASS LAB
- 250-NONCLASS LAB
- 255-NONCLASS LAB SERVICE
- 310-OFFICE
- 315-OFFICE SERVICE
- 317-DEPARTMENT COMMONS
- 350-CONFERENCE
- 410-STUDY
- 455-STUDY SERVICE
- 630 - DINING
- 635 - FOOD SERVICE
- 650-LOUNGE
- 651-BUILDING COMMONS
- 680-MEETING ROOM
- 685 - MEETING ROOM SERVICE
- 730-CENTRAL STORAGE
- Calculating...

OPTION F
THIRD FLOOR USE

Conceptual Aerial of South Facade and Plaza - Option F UPDATED RENDERING IN PROGRESS



Conceptual South Facade and Plaza - Option F UPDATED RENDERING IN PROGRESS



Conceptual Plaza Redesign - Option F UPDATED RENDERING IN PROGRESS





AMERICAN STRUCTUREPOINT INC.

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REITANO DESIGN GROUP

BLUNDALL ASSOCIATES, INC.